▧ CUET Computer Networks (CSE-311/411) — Complete 5-Year Solutions (2019–2023)

Every unique question from the last 5 years, solved once. Repeated questions are merged.
Sources: Rifat PDF (page refs), materials/slides/, materials/books/

PART A: NETWORK BASICS & PHYSICAL LAYER

Q: OSI vs TCP/IP Model — Compare and Contrast

Asked: 2023 Q.1a, 2022 Q.2a (layers), 2021 Q.2a (TCP/IP layers), 2020 Q.2a, 2019 Q.1a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2023 Q.1a Model Answer
Difference Between OSI and TCP/IP Reference Model:

Feature	OSI Reference Model	TCP/IP Reference Model
Layer Count	Defines 7 discrete layers (Application, Presentation, Session, Transport, Network, Data Link, Physical).	Defined with 4 abstraction layers (Application, Transport, Internet, Network Access).
Layer Grouping	Upper layers (Application, Presentation, Session) handle software; lower layers handle network hardware and routing.	Combines OSI layers 5, 6, 7 into a single Application layer, and layers 1, 2 into a Network Access layer.
Development	Theoretical model developed by ISO to standardize communication standards.	Practical, deployed model developed by DARPA/DoD around actual protocols (TCP and IP).
Connection Service	The Network layer supports both connectionless and connection-oriented communication.	The Internet layer exclusively supports connectionless communication (IP).
Protocol Independence	Model was defined before protocols, making it very protocol-independent.	Model is tightly coupled with its protocols; it's a description of existing protocols.

Justification: "Encapsulation helps to form the layer"

- **Definition:** Encapsulation is the process in which each OSI layer receives data from the layer above, treats it as a payload, and adds its own protocol-specific control information (a header or trailer).
- **Forming the Layer:** The OSI layers use these various forms of control information to communicate with their peer layers in other computer systems. For example, a TCP segment at the Transport layer is encapsulated inside an IP packet at the Network layer.
- **Modularity:** Because each layer is abstracted and only reads/processes its own specific header, encapsulation inherently forms strict layer boundaries, ensuring that a protocol at one layer does not need to understand the protocols at another layer.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1, 4 | Slides: OSI_and_TCP_IP_Layering_Models.pdf

Feature	OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Development	Theoretical model by ISO, defined before protocols	Practical model by DoD, built around existing protocols
Layers	Application, Presentation, Session, Transport, Network, Data Link, Physical	Application, Transport, Internet, Network Access
Network Layer	Supports both connectionless and connection-oriented	Supports only connectionless (IP)
Transport Layer	Supports only connection-oriented	Supports both TCP (connection) and UDP (connectionless)
Protocol Coupling	Protocol-independent	Tightly coupled with TCP/IP suite

Why TCP/IP is more popular: It was built organically alongside real networks (ARPANET/UNIX). OSI was academic and arrived too late. TCP/IP's 4-layer simplicity maps directly to how software is programmed.

Encapsulation forms layers: each layer receives data from above, treats it as payload, adds its own header. This creates strict boundaries — a Transport protocol need not understand Network layer internals.

Q: What is Computer Network? Components? Internet as "Network of Networks"

Asked: 2021 Q.1a, 2020 Q.1a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.1a Model Answer**
> **Computer Network:**
> A computer network is a structurally interconnected collection of autonomous computers and devices that share resources (like printers or internet access) and exchange information using standard communication protocols over digital interconnections (wired or wireless).

Distinctions:
(i) Computer vs. Autonomous Systems (AS):

- **Computer:** A single functional end-device (host) capable of computation and communication on a network.
- **Autonomous System (AS):** defined strictly as a set of routers under a single technical administration. It uses an Interior Gateway Protocol (IGP) to route packets within the AS, and presents a single unified routing policy to the global Internet.

(ii) Client-Server Model vs. Peer-to-Peer Model:
(Discussed fully in 2023 Q.4a. Briefly: Client-Server is centralized where thin clients rigidly request data from powerful central servers. Peer-to-Peer is decentralized where every node acts interchangeably as both a client and a server, directly sharing resources continuously).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1–3

Computer Network: A collection of autonomous computers interconnected by a single technology, sharing resources using standard protocols.

Components: (1) End Devices (PCs, phones), (2) Network Devices (switches, routers), (3) Transmission Media (twisted pair, fiber, wireless), (4) Protocols (TCP/IP, Ethernet).

Internet = "Network of Networks": The Internet is not one network — it interconnects thousands of independent LANs, WANs, and ISPs using TCP/IP as the common glue.

Q: Network Topologies — Types, Pros, Cons

Asked: 2023 Q.1c/Q.2a, 2022 Q.2a, 2021 Q.1b, 2020 Q.1b, 2019 Q.1b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.1c Model Answer**
> **Hybridizing Star and Ring Topologies:**
> When these topologies are hybridized, the physical network is wired like a **Star** (with all nodes connecting to a central hub/coupler), but logically the data travels in a **Ring** path. The central device acts as a Multistation Access Unit (MAU) that internally bridges the transmit pair of one node to the receive pair of the next node.

Circumstances and Challenges in Setup:

- **Cabling Medium:** Fiber optics or twisted pair are suitable for use in ring and star topologies. Setting this up requires running point-to-point links (one for transmit, one for receive) from every node back to the central hub.
- **Central Point of Failure:** While the logical ring prevents packet collisions, the physical setup creates a single point of failure at the central hub. If the internal ring wiring in the MAU fails, the entire network drops.
- **Fault Detection:** Setting up the network requires hardware that can automatically bypass a faulty node or broken cable run, allowing the ring to instantly close loop and preserve the remaining connectivity.

(See diagram in the final document: Network Topologies)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 6, 11, 14, 15 | Slides: Network_Topology_Overview.pdf, LAN_Technology_and_Network_Topologies.pdf

Topology	Pros	Cons
Bus	Simple, least cabling	Backbone break kills all; heavy collisions
Star	Robust (one cable break = one PC down); easy troubleshooting	Central hub = single point of failure; intensive cabling
Ring	Token passing = zero collisions; fair access	Ring break halts entire network
Mesh	Highly fault-tolerant; no single point of failure	Expensive: n(n-1)/2 links needed
Tree/Hybrid	Scalable; hierarchical grouping	Complex; inherits star's central-point weakness

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Q: Protocols and Topologies Definitions; Logical Topologies

Asked: 2022 Q.2a, 2019 Q.1b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.2a Model Answer**
> **Definitions:**

- **Physical Topology:** Refers to the physical layout and actual geometric arrangement of nodes, cables, and switches in a network.
- **Logical Topology:** Refers to how data actually flows and is transmitted through the network from one device to the next, regardless of the physical layout (e.g., passing a token logically in a ring over a physical star wiring).

Advantages and Disadvantages of Topologies:

- **Bus Topology:**
  - **Advantages:** Simple layout. Easy to connect a computer or peripheral to a linear bus. Requires less cable length than a star topology.
  - **Disadvantages:** Entire network shuts down if there is a break in the main backbone cable. Terminators are required at both ends. Difficult to identify the problem if the entire network goes down.
- **Ring Topology:**
  - **Advantages:** Every node gets equal access to the token (predictable delay). Performs better than a bus topology under a heavy network load.
  - **Disadvantages:** A break in the closed ring cable or the failure of a single node can bring down the entire network (unless special dual-ring bypass hardware is used). Difficult to add or remove nodes without disrupting the active ring.
- **Star Topology:**
  - **Advantages:** High speed data transfer; particularly when the star coupler acts as a switch. Easiest to maintain and troubleshoot. Very flexible. A failure of one node or link does not affect the rest of the network.
  - **Disadvantages:** Complete dependency on the central node/coupler; if it fails, the entire network drops. Cabling costs are higher because the number of links is strictly proportional to n (every node needs an independent cable).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Protocol:** A set of rules governing data format, timing, sequencing, and error control.
- **Physical Topology:** Actual cable layout. **Logical Topology:** How data flows conceptually.
- **Logical Ring (Token Ring):** Deterministic, zero collisions, fair. Slow token overhead.
- **Logical Bus (Ethernet CSMA/CD):** Fast under light loads; collisions surge under heavy load.

Q: Unicast, Broadcast, Multicast

Asked: 2020 Q.1b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2020 Q.1b Model Answer**
> **(i) Modes of Communication:**

- **Unicast:** Communication is strictly one-to-one. A single sender transmits a message directed exclusively to a single, specific receiver MAC/IP address.
- **Multicast:** Communication is strictly one-to-many. A sender transmits a single message that is intelligently routed only to a specific, subscribed group of receivers (e.g., IPTV).
- **Broadcast:** Communication is strictly one-to-all. A sender transmits a message meant to be received and processed by absolutely every single device physically present on the local subnet.

(ii) Comparison of Network Topologies:
(Summarized from previous detailed answers)

- **Bus:**
  - *Pros:* Simple layout, requires the least amount of cabling.
  - *Cons:* Entire network crashes if the central coaxial backbone breaks. Heavy collisions.
- **Star:**
  - *Pros:* Highly robust; if one cable breaks, only that single PC drops. Excellent for unguided media and modern Ethernet.
  - *Cons:* A complete central point of failure (if the main switch dies, everything dies). Requires intensive cabling.
- **Ring:**
  - *Pros:* Token passing naturally prevents all data collisions. Fair access for all nodes.
  - *Cons:* Physically breaking the closed ring cable immediately halts all communication.
- **Mesh:**
  - *Pros:* Highly reliable and vastly fault-tolerant (no single point of failure).
  - *Cons:* Exceptionally high completely wiring cost and rigid physical complexity ($n(n-1)/2$ links required).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Unicast:** One-to-one (single sender to single receiver).
- **Broadcast:** One-to-all (every device on the subnet receives it). Example: ARP Request.
- **Multicast:** One-to-many (only subscribed group members receive). Example: IPTV.

Q: Wired vs Wireless Networking

Asked: 2021 Q.1c, 2020 Q.1c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2021 Q.1c Model Answer
(i) Wired vs. Wireless Networking:

- **Wired:** Uses physical guided media (Cat6 Copper, Fiber optics). Guarantees ultra-high strict bandwidth (up to 100 Gbps+ on fiber), severe low latency, and superb physical security against hacking. Lacks physical mobility.
- **Wireless:** Uses unguided electromagnetic waves (Wi-Fi, 5G). Allows infinite geographical mobility and fast ad-hoc setup. Suffers from signal attenuation, interference, and demands heavy encryption overhead (WPA3) to secure the open air waves.

(ii) Collision Domain vs. Broadcast Domain:

- **Collision Domain:** A localized segment where frames will collide if sent simultaneously. Hubs form one large collision domain. Switches filter packets, forwarding frames only to the necessary segments, meaning switch ports become separate, isolated collision domains.
- **Broadcast Domain:** A broader segment where a Layer 2 Broadcast safely reaches every host. Switches forward broadcasts across all ports. Only Layer 3 Routers block broadcasts.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 58–59

- **Wired:** Guided media (Cat6, Fiber). High bandwidth (100 Gbps on fiber), low latency, physically secure. No mobility.
- **Wireless:** Unguided (Wi-Fi, 5G). Full mobility. Suffers attenuation, interference, needs encryption (WPA3).

Q: Collision Domain vs Broadcast Domain

Asked: 2023 Q.1c, 2021 Q.1c, 2020 Q.1c, 2019 Q.5a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

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Circumstances and Challenges in Setup:

- **Cabling Medium:** Fiber optics or twisted pair are suitable for use in ring and star topologies. Setting this up requires running point-to-point links (one for transmit, one for receive) from every node back to the central hub.
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- **Fault Detection:** Setting up the network requires hardware that can automatically bypass a faulty node or broken cable run, allowing the ring to instantly close loop and preserve the remaining connectivity.

(See diagram in the final document: Network Topologies)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 61–67, 72 | Slides: Hardware_Hub_Bridge_Router_Switch_QA.pdf

- **Collision Domain:** Segment where simultaneous transmissions collide. Hubs = one big domain. Each switch port = separate domain.
- **Broadcast Domain:** Segment where a Layer-2 broadcast reaches all hosts. Switches forward broadcasts. Only routers block them.
- **Situation where broadcast is necessary:** ARP — a host broadcasts "Who has IP 192.168.1.5?" to discover the MAC address.

Q: Hub, Switch, Router — Construction and Operation

Asked: 2021 Q.1d, 2020 Q.1c/d, 2019 Q.8c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.1d Model Answer**
> **(i) Hub, Switch, and Router:**

- **Hub (Layer 1 - Physical):** A "dumb" multiport repeater. It rigidly receives a signal on one port and blindly broadcasts it out of completely all other ports. It creates massive single collision and broadcast domains.
- **Switch (Layer 2 - Data Link):** An intelligent bridging device. It actively learns MAC addresses and builds a MAC Table. When it receives a frame, it inspects the destination MAC and structurally forwards the frame *only* out of the specific port belonging to that destination. It eliminates collisions.
- **Router (Layer 3 - Network):** Determines the mathematically shortest overarching path between completely different distinct IP networks. It rigidly breaks broadcast domains, aggressively reads IP destination addresses, and utilizes routing tables to safely bounce packets across the internet grid.

(ii) Token Passing and FDDI:

- **Token Passing:** A deterministic channel access method used in Ring networks. A small specialized frame called the "Token" circulates the wire endlessly. A station can purely transmit data *only* if it physically captures the free Token. This structurally prevents all collisions.
- **FDDI (Fiber Distributed Data Interface):** Extremely robust standard for data transmission on fiber optic lines in LANs. It fundamentally uses a dual-ring (primary and secondary) rotating token-passing architecture. If a fiber cable gets cut, the dual rings instantly fold into a single operational U-turn ring, making FDDI incredibly fault-tolerant against catastrophic physical cable cuts.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 14, 62, 72 | Slides: Computer_Networking_Devices_Overview.pdf

- **Hub (Layer 1):** Dumb multiport repeater. Broadcasts to ALL ports. Creates one collision domain.
- **Switch (Layer 2):** Learns MAC addresses, forwards frames only to the correct port. Eliminates collisions.
- **Router (Layer 3):** Routes between different IP networks using routing tables. Breaks broadcast domains.

Q: Token Passing and FDDI

Asked: 2021 Q.1d, 2019 Q.8b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

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[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 73 | Slides: LAN_Technologies_Must_Read.pdf

- **Token Passing:** Deterministic access method. A "Token" frame circulates the ring. Only the token holder can transmit. Zero collisions by design.
- **FDDI (Fiber Distributed Data Interface):** 100 Mbps fiber-optic LAN standard using a dual counter-rotating ring with token passing. If a cable is cut, the two rings "wrap" into a single loop — providing fault tolerance.

Q: Shannon's Theorem & Nyquist Theorem

Asked: 2023 Q.2b/c, 2019 Q.2d, CSE-411 Q.3c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.2b Model Answer**
> **Using the Shannon Theorem:**

- Yes, the **Shannon capacity gives us the upper theoretical limit** (maximum bit rate) for a channel. It defines the theoretical max bit rate in a noisy channel where random noise is present.
- Formula: `C = B log₂(1 + SNR)`. Even though Shannon formula gives this upper limit, for better performance, practical transmission usually chooses a target bit rate lower than `C` and uses the Nyquist formula to find the required signal levels.

How SNR differs from Channel Bandwidth:

- **Channel Bandwidth (B):** Measured in Hertz (Hz), it represents the physical frequency range that the medium can reliably transmit (e.g., 300 to 3300 Hz for a telephone line). It is a property of the physical medium.
- **Signal-to-Noise Ratio (SNR):** Measured in Decibels (dB), it is the ratio of average signal power to average noise power (S/N). It represents the clarity or quality of the signal against background thermal noise, not the physical frequency range.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 23 | Slides: Nyquist_and_Shannon_Theorems.pdf, Shannon_Channel_Capacity.pdf

Nyquist (Noiseless): $C = 2 \times B \times \log_2(L)$ where B=bandwidth, L=signal levels.
Shannon (Noisy): $C = B \times \log_2(1 + SNR)$ — gives the theoretical maximum.

Solved: Bandwidth = 4300 Hz (2019 Q.2d: C=8600, L=2 → B = 8600/2 = 4300 Hz)
Solved: Bandwidth = 2000 Hz (2019 Q.2d variant: C=4000, S/N=3 → B = 4000/$\log_2$(4) = 2000 Hz)
Solved: C ≈ 10,656 bps (CSE-411: B=1600, SNR=20dB → SNR_linear=100, C=1600×$\log_2$(101)≈10,656)

PART B: DATA LINK LAYER

Q: Framing — Definition and Methods

Asked: 2023 Q.3a, 2021 Q.2a, 2020 Q.2b, 2019 Q.2b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.3a Model Answer**
> **Bit Stuffing vs. Character Stuffing:**

- **Framing Boundary Context:** The Data Link Layer translates the raw bit stream into discrete units called frames. To know where a frame begins and ends, unique flag sequences are used.
- **Character Stuffing:** Used in byte-oriented protocols. A special byte (flag character) marks the boundary. If the flag character appears in the data payload, an 'escape' character is inserted (stuffed) before it to ensure it isn't misread as the end of the frame.
- **Bit Stuffing:** Used in bit-oriented protocols (like HDLC). The frame boundary is defined by the bit sequence `01111110`. If the user data contains five consecutive `1`s, the sender's hardware automatically inserts (stuffs) a `0` immediately after them to prevent it from accidentally creating the flag sequence. The receiver detects the five `1`s and automatically removes the stuffed `0`.

Framing Process:

1. **Header:** Contains control information (source/destination MAC addresses).
2. **Payload:** The physical layer's raw bit stream (upper layer packet). Stuffing is applied here to prevent accidental flag sequences.
3. **Trailer:** Contains Error Control (e.g., CRC) to detect single-bit or burst errors.
4. **Flags:** Unique sequences (e.g., `01111110`) placed at the start and end of the frame.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 6, 8, 32–35 | Slides: Data_Link_Framing_and_Error_Control.pdf

Framing: The Layer-2 process of breaking a raw bit stream into discrete logical "frames" so the receiver knows where a message starts and ends.

Four Methods: (1) Character Count, (2) Character/Byte Stuffing, (3) Bit Stuffing, (4) Physical Layer Coding Violations.

- **Character Count:** Frame length embedded in header. Simple but brittle — a single bit error in the count field loses sync permanently.
- **Byte Stuffing:** Special FLAG byte marks boundaries. If FLAG appears in data, an ESC byte is stuffed before it.
- **Bit Stuffing:** Flag = `01111110`. If five consecutive `1`s appear in data, a `0` is stuffed after them. Receiver removes it automatically.

Destuffing Example (2023 Q.3b): Data `0110 0111 1100 1111 0111 1101`
→ Find five `1`s followed by `0`: remove the stuffed `0` after sequences of five `1`s.

Q: Flow Control — Sliding Window Protocol

Asked: 2023 Q.3b, 2021 Q.2c, 2020 Q.3a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.3b Model Answer**
> **Advantage of Go-Back-N over Stop-&-Wait:**

- **Stop-and-Wait ARQ:** Very inefficient because it transmits one frame, and then stops to wait fully for the propagation delay of the frame going and the ACK coming back. If the line is long or high-bandwidth, the channel is mostly idle.
- **Go-Back-N ARQ (Pipeline):** Allows the sender to transmit multiple frames consecutively before requiring an acknowledgment. This keeps the network pipe full and drastically increases channel utilization and overall throughput, making it highly advantageous over Stop-and-Wait.

Window Size and Propagation Delay:

- **What Window Size Specifies:** The window size (`N`) specifies the maximum number of unacknowledged frames the sender is allowed to have in transit at any given time.
- **Relation to Propagation Delay:** The window size must be mathematically related to the round-trip propagation delay. **Propagation Delay** refers to the time necessary for a single bit to travel end-to-end on a physical wire. To maximize efficiency (so the sender doesn't stall waiting for ACKs), the window size `N` must be large enough to hold all the frames that can be transmitted during the Round Trip Time (RTT), preventing the sender from ever entering an forced idle phase.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 54, 55, 119 | Slides: Data_Link_Flow_and_Error_Control.pdf

Why Sliding Window? Stop-and-Wait is inefficient — the sender idles 90% waiting for ACKs. Sliding Window allows sending W frames before needing an ACK, keeping the pipe full.

Go-Back-N vs Selective Repeat:

- **Go-Back-N:** Receiver rejects out-of-order frames. If frame 3 is lost, frames 4,5,6 are discarded and sender re-sends 3,4,5,6. Simple but wasteful.
- **Selective Repeat:** Receiver buffers out-of-order frames. Only the lost frame is retransmitted. Bandwidth efficient but complex buffering.

Q: CSMA/CD and CSMA/CA

Asked: 2023 Q.1c, 2022 Q.3b, 2021 Q.2a, 2020 Q.3d, 2019 Q.1c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.1c Model Answer**
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Circumstances and Challenges in Setup:

- **Cabling Medium:** Fiber optics or twisted pair are suitable for use in ring and star topologies. Setting this up requires running point-to-point links (one for transmit, one for receive) from every node back to the central hub.
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(See diagram in the final document: Network Topologies)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 61 | Slides: Ethernet_CSMACD_Exponential_Backoff.pdf, Wireless_MAC_Protocols.pdf

- **CSMA/CD (Collision Detection — Wired Ethernet):** Listen → Transmit → Monitor voltage. If spike detected (collision) → Abort → Send Jam Signal → Binary Exponential Backoff → Retry.
- **CSMA/CA (Collision Avoidance — Wireless 802.11):** Cannot detect collisions while transmitting. Uses RTS/CTS handshake to reserve the channel before sending.

1-persistent vs Non-persistent CSMA:

- **1-persistent:** If channel busy, keep listening; transmit immediately when idle. Aggressive — high collision risk.
- **Non-persistent:** If busy, wait a random time before listening again. Reduces collisions but adds latency.

Q: CRC Checksum Calculations

Asked: 2023 Q.3a, 2022 Q.2c, 2020 Q.2d, 2019 Q.2c, CSE-411 Q.3a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

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3. **Trailer:** Contains Error Control (e.g., CRC) to detect single-bit or burst errors.
4. **Flags:** Unique sequences (e.g., `01111110`) placed at the start and end of the frame.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 21, 135, 140 | Slides: Packets_Frames_and_Error_Detection.pdf

Solved: Frame `101101110011`, $G(X) = X^4+X^2+1$ → Binary `10101`

1. Append 4 zeros: `1011011100110000`
2. Modulo-2 divide by `10101`
3. Remainder = CRC checksum → Append to original frame

Solved: Frame `101100101`, $G(x) = x^3+x^2+1$ → Binary `1101`

1. Append 3 zeros: `101100101000`
2. Modulo-2 division:

```
          111101100
     ________________
1101 | 101100101000
       1101
       ----
        1100
        1101
        ----
          0110
          0000
          ----
           1101
           1101
           ----
             0001
             0000
             ----
               0010
               0000
               ----
                 0100
                 0000
                 ----
                  1000
                  1101
                  ----
                   101 (Remainder)
```

Transmitted Frame: `101100101101`

Solved: Frame `10101011011`, Generator `10001` — Verify reception of `10101011010`

Divide received `10101011010` by `10001`. Remainder = `1010` ≠ 0. **NOT accurate reception — errors occurred.**

Solved: Data `10011011100`, $G(x) = x^4+x^2+1$ → `10101`

Append 4 zeros → `100110111000000`. After division, Remainder = `0100`.
Transmitted Codeword: `100110111000100`

Q: Hamming Code — Error Correction

Asked: 2023 Q.2c ("CAFE"), 2021 Q.2d ("SEE"), 2020 Q.3c ("CSE"), 2019 Q.4c ("CUET"), CSE-411 (data 1010)

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.2c Model Answer**
> **Nyquist Bit Rate:**

- **Definition:** The Nyquist theorem defines the theoretical maximum bit rate in a **noiseless (perfect)** channel.
- It states that even perfect channels have limited capacity. To find what data rate and signal levels are appropriate, the formula used is: `BitRate = 2 * B * log₂(L)`, where `B` is bandwidth and `L` is the number of signal levels.

Relationship Between Shannon and Nyquist:

- **Shannon gives the theoretical limit (Capacity):** Shannon's theorem extends Nyquist by accounting for random noise. In any real channel, the transmission limit cannot exceed the Shannon Capacity `C = B log₂(1 + SNR)`, no matter how many signal levels `L` are used.
- **Nyquist gives the required signaling levels:** In practice, both methods are used together. First, the Shannon capacity gives us the upper limit based on the noise. Then, the Nyquist formula tells us how many signal levels (`L`) we need to actually achieve a target data rate that is safely below the Shannon limit.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 42, 46 | Slides: Data_Link_Framing_and_Error_Control.pdf

Method: For 7 data bits → 4 parity bits (positions 1,2,4,8). Total = 11 bits.
Format: P_1, P_2, D_3, P_4, D_5, D_6, D_7, P_8, D_9, D_{10}, D_{11}

Letter	ASCII (7-bit)	Hamming Code (11-bit)
C (1000011)	D=1,0,0,0,0,1,1	`01100000011`
A (1000001)	D=1,0,0,0,0,0,1	`11100000001`
F (1000110)	D=1,0,0,0,1,1,0	`01100000110`
E (1000101)	D=1,0,0,0,1,0,1	`10100000101`
S (1010011)	D=1,0,1,0,0,1,1	`00110100011`
U (1010101)	D=1,0,1,0,1,0,1	`11110100101`
T (1010100)	D=1,0,1,0,1,0,0	`00110101100`

Solved: 4-bit data `1010` (CSE-411)

3 parity bits needed. Format: P_1,P_2,D_3,P_4,D_5,D_6,D_7. Data: D_3=1, D_5=0, D_6=1, D_7=0.
P_1(1,3,5,7)=1, P_2(2,3,6,7)=0, P_4(4,5,6,7)=1. **Codeword:** `1011010`
Error correction: If bit 5 flips → Syndrome = $P_4P_2P_1$ = 101_2 = 5 → flip bit 5 back.

PART C: TRANSPORT LAYER

Q: TCP vs UDP

Asked: 2023 Q.3c, 2022 Q.3a, 2021 Q.3a, 2020 Q.2c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.3c Model Answer**
> **Technique Specified:**
> This specifies the **Cyclic Redundancy Check (CRC)** error detection technique, utilizing polynomial arithmetic done modulo 2 (which is effectively XOR arithmetic without carries).

Given:

- Message (`M`): `1101`
- Message appended with 3 zeros for division (`k` as per lecture): `1101000`
- Generator Polynomial / Divisor (`P`): $x^3 + x + 1$ which translates to binary `1011`. (Since we appended 3 zeros, the divisor must be 4 bits).

Calculation (Modulo-2 Division):
We divide `1101000` by `1011` using XOR:

```
         1110  (Quotient)
     ________
1011 | 1101000
       1011
       ----
        1100
        1011
        ----
         1110
         1011
         ----
          1010
          0000
          ----
           101  (Remainder) ---> Actually, let's recalculate precisely:
```

Precise Modulo-2 XOR Steps:

1. `1101` XOR `1011` = `0110`
2. Bring down `0` -> `1100`
3. `1100` XOR `1011` = `0111`
4. Bring down `0` -> `1110`
5. `1110` XOR `1011` = `0101`
6. Bring down `0` -> `1010`
7. `1010` XOR `1011` = `0001`

- **Remainder (r):** `001`

Final Data Sent:
The CRC remainder `001` is appended to the original message `1101`.
The bit frame `(k+r)` actually transmitted is **`1101001`**.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 116–120 | Slides: Transport_Protocols_UDP_TCP_RTP.pdf

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed ordered delivery via ACKs	Best-effort, unreliable
Flow/Congestion	Sliding window + AIMD	None
Header Size	20 bytes minimum	8 bytes
Use Cases	FTP, HTTP, Email	DNS, VoIP, Gaming, Streaming

Q: TCP 3-Way Handshake & Congestion Control

Asked: 2022 Q.3a, 2021 Q.3a, 2020 Q.2c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.3a Model Answer**
> **TCP (Transmission Control Protocol):**
> TCP is the primary transport level protocol of the Internet Protocol Suite that provides reliable, byte-oriented stream management. It offers a connection-oriented service where applications interact with TCP to send and receive data reliably across an IP network with sequencing and flow control.

TCP 3-Way Handshake Process (Connection Establishment):
The reliable handshake ensures both communication entities establish parameters before sending data.

1. **SYN (Step 1):** The Active Participant (Client) sends a segment with the SYN (Synchronize) flag set to request a connection to the Passive Participant (Server). It includes the client's initial Sequence number `X`.
2. **SYN-ACK (Step 2):** The Server receives the SYN, allocates data structures, and replies with a segment that has *both* the SYN flag and ACK flag set. The Server provides its own initial Sequence number `Y`, and acknowledges the client's sequence by setting the Acknowledgment number to `X + 1`.
3. **ACK (Step 3):** The Client receives the SYN-ACK, and sends a final segment with just the ACK flag set, acknowledging the server's sequence by setting the Acknowledgment number to `Y + 1`.

(See diagram in the final document: TCP 3-Way Handshake)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 119, 122 | Slides: Transmission_Control_Protocol_TCP.pdf

3-Way Handshake: (1) Client sends **SYN** (seq=X). (2) Server replies **SYN-ACK** (seq=Y, ack=X+1). (3) Client sends **ACK** (ack=Y+1). Connection established.

Congestion Control (4 phases):

1. **Slow Start:** cwnd=1 MSS, doubles each RTT until threshold.
2. **Congestion Avoidance (AIMD):** After threshold, cwnd increases by +1 per RTT.
3. **Fast Retransmit:** 3 duplicate ACKs → retransmit lost segment immediately.
4. **Fast Recovery:** Halve threshold, resume from congestion avoidance (not slow start).

Q: Segment vs Packet vs Frame; Port vs Socket

Asked: 2020 Q.3a/b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2020 Q.3a Model Answer**

- **Segment vs. Packet vs. Frame:**
  - **Segment:** The payload strictly constructed at Layer 4 (Transport, TCP). Does not use IP headers.
  - **Packet (Datagram):** Built strictly at Layer 3 (Network, IP). It encapsulates the TCP segment by adding Source and Destination IP Addresses.
  - **Frame:** Built strictly at Layer 2 (Data Link). It rigidly encapsulates the IP Datagram by adding physical MAC addresses and a CRC error-correction trailer.
- **Port vs. Socket:**
  - **Port:** A 16-bit software construct strictly identifying a specific application process running inside a host computer (e.g., HTTP uses Port 80).
  - **Socket:** The exact combination of an IP address physically paired with a Port number (e.g., `192.168.1.1:80`). It represents a definitive mathematical endpoint of a two-way communication link.
- **Sliding Window:** *(Detailed in 2021 Q.2c).*

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 32, 116

- **Segment:** Layer 4 (Transport/TCP). Contains port numbers.
- **Packet/Datagram:** Layer 3 (Network/IP). Contains source/destination IP.
- **Frame:** Layer 2 (Data Link). Contains source/destination MAC + CRC trailer.
- **Port:** 16-bit number identifying a specific application (e.g., HTTP=80).
- **Socket:** IP Address + Port Number (e.g., `192.168.1.1:80`). Unique communication endpoint.

PART D: NETWORK LAYER

Q: IP Addressing — Classes, Classful vs Classless, Public vs Private

Asked: 2023 Q.4a/b, 2022 Q.3b/Q.4a, 2020 Q.4a, 2019 Q.4a–d

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.4a Model Answer**
> **IP Address Classes and Ranges:**
> Traditional IPv4 addresses are divided into classes to accommodate networks of varying sizes.

- **Class A:** `0.0.0.0` to `127.255.255.255`. Designed for extremely large networks.
- **Class B:** `128.0.0.0` to `191.255.255.255`. Designed for medium-sized networks.
- **Class C:** `192.0.0.0` to `223.255.255.255`. Designed for smaller local area networks.
- **Class D:** `224.0.0.0` to `239.255.255.255`. Reserved strictly for Multicast addressing.
- **Class E:** `240.0.0.0` to `255.255.255.255`. Reserved for experimental/future use.

Components to Determine Identifiers:
An IP address is logically broken into two primary components, separated using a subnet mask or CIDR notation:

1. **Network Prefix (Network ID):** A common group of most-significant bits used for routing datagrams to the correct specific network.
2. **Rest Field (Host ID):** The remaining bits used to uniquely identify a specific host/node within that localized network.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 103–110 | Slides: IP_Addressing_Main_Slide.pdf, IPv4_Addresses_Classful_Addressing.pdf

Class	Range	Default Mask	Purpose
A	0.0.0.0 – 127.255.255.255	/8	Very large networks
B	128.0.0.0 – 191.255.255.255	/16	Medium networks
C	192.0.0.0 – 223.255.255.255	/24	Small networks
D	224.0.0.0 – 239.255.255.255	—	Multicast

- **Classful:** Fixed class boundaries. **Classless (CIDR):** Variable-length prefix (e.g., /22).
- **Public IP:** Globally routable. **Private IP:** 10.x.x.x, 172.16–31.x.x, 192.168.x.x (not routable).
- **Subnet Mask:** 1s = network, 0s = host. **Wildcard Mask:** Inverse (used in ACLs).

Q: Subnetting & VLSM Calculations

Asked: 2023 Q.4c, 2020 Q.4b, 2019 Q.5b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.4c Model Answer**
> **Step 1: Identify the Base Network Block**
> The range `172.16.0.0` to `172.16.3.255` encompasses 1024 IP addresses.

- Network: `172.16.0.0/22`
- In VLSM (Variable Length Subnet Masking), we allocate the largest requirement first.

Step 2: Allocate Subnet 1 (Requirement: 50 PCs)

- Address space needed: 50 hosts + 2 (Network & Broadcast address) = 52 addresses.
- The closest power of 2 that supports this is $2^6 = 64$. Thus, we need 6 bits for the host ID.
- **Prefix length:** 32 - 6 = **`/26`**
- **Subnet Mask:** `255.255.255.192`
- **Network Address:** `172.16.0.0/26`
- **Range:** `172.16.0.0` to `172.16.0.63`
  - *First Usable:* `172.16.0.1`
  - *Last Usable:* `172.16.0.62`
  - *Broadcast:* `172.16.0.63`

Step 3: Allocate Subnet 2 (Requirement: 23 PCs)

- Pick up from the next available continuous address block: `172.16.0.64`.
- Address space needed: 23 hosts + 2 = 25 addresses.
- The closest power of 2 that supports this is $2^5 = 32$. Thus, we need 5 bits for the host ID.
- **Prefix length:** 32 - 5 = **`/27`**
- **Subnet Mask:** `255.255.255.224`
- **Network Address:** `172.16.0.64/27`
- **Range:** `172.16.0.64` to `172.16.0.95`
  - *First Usable:* `172.16.0.65`
  - *Last Usable:* `172.16.0.94`
  - *Broadcast:* `172.16.0.95`

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 106–109 | Slides: Subnetting_Supernetting_Classless_Addressing.pdf, Network_Season_07_VLSM.pdf

Formula: Usable hosts = 2^h − 2. Find smallest h where 2^h − 2 ≥ requirement.

Solved: 172.16.2.1/22 for 50 PCs and 23 PCs (VLSM)

- **50 PCs:** Need 64 addresses (2^6). Prefix = /26, Mask = .192. Network: 172.16.0.0/26. Usable: .1–.62, Broadcast: .63
- **23 PCs:** Need 32 addresses (2^5). Prefix = /27, Mask = .224. Network: 172.16.0.64/27. Usable: .65–.94, Broadcast: .95

Solved: 210.210.210.0/24 for min 15 usable hosts

Need 2^5=32 (since 2^4−2=14 fails). Borrow 3 bits → /27. Creates 8 subnets of 30 usable hosts each.

Solved: 100.100.100.0/24 for min 30 usable hosts

Need 2^5=32 (30 usable). Borrow 3 bits → /27. Creates 8 subnets.

Q: IPv4 vs IPv6

Asked: 2023 Q.4b, 2022 Q.5b, 2020 Q.4c

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.4b Model Answer**
> > **Subnetting:**
>
> - **Definition:** Subnetting is the process of dividing a single IP network into smaller, logical sub-divisions called subnets.
> - **How it works:** In subnetting, bits from the Host ID field are "borrowed" to be used as a subnet identifier. For example, breaking a `/24` network into smaller `/26` sub-networks. This isolates traffic and improves security.
>
> **Supernetting (Route Aggregation):**
>
> - **Definition:** Supernetting is the process of combining several smaller IP networks into one large network with a common Classless Inter-Domain Routing (CIDR) prefix.
> - **How it works / Difference:** It is the exact inverse of subnetting. In supernetting, bits from the Network ID are "borrowed" to be used as the Host ID. For example, combining networks `192.60.2.0/24` and `192.60.3.0/24` into a single supernetwork `192.60.2.0/23`.
> - **Purpose:** Supernetting drastically reduces the number of entries in a routing table, simplifying the routing process and conserving router memory (aggregation).

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 59, 102, 112 | Slides: IP_Addressing_IPv4_and_IPv6.pdf

Feature	IPv4	IPv6
Address Size	32-bit (4.3 billion)	128-bit (virtually infinite)
Header	Variable (20+ bytes with options)	Fixed 40-byte base header
Fragmentation	Routers can fragment	Only endpoints fragment (Path MTU Discovery)
NAT	Required (address exhaustion)	Not needed

Q: Subnetting vs Supernetting

Asked: 2023 Q.4b, 2022 Q.4b, 2020 Q.4a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.4b Model Answer**
> > **Subnetting:**
>
> - **Definition:** Subnetting is the process of dividing a single IP network into smaller, logical sub-divisions called subnets.
> - **How it works:** In subnetting, bits from the Host ID field are "borrowed" to be used as a subnet identifier. For example, breaking a `/24` network into smaller `/26` sub-networks. This isolates traffic and improves security.
>
> **Supernetting (Route Aggregation):**
>
> - **Definition:** Supernetting is the process of combining several smaller IP networks into one large network with a common Classless Inter-Domain Routing (CIDR) prefix.
> - **How it works / Difference:** It is the exact inverse of subnetting. In supernetting, bits from the Network ID are "borrowed" to be used as the Host ID. For example, combining networks `192.60.2.0/24` and `192.60.3.0/24` into a single supernetwork `192.60.2.0/23`.
> - **Purpose:** Supernetting drastically reduces the number of entries in a routing table, simplifying the routing process and conserving router memory (aggregation).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Subnetting:** Borrow host bits → divide one network into smaller subnets.
- **Supernetting (Route Aggregation):** Borrow network bits → combine multiple networks into one supernetwork. Reduces routing table size.

Q: CIDR Routing Table Lookup

Asked: 2022 Q.5a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.5a Model Answer

Feature	Connection-Oriented Service	Connection-Less Service
Analogy	Analogous to the telephone system.	Analogous to the postal system.
Setup Phase	Requires communication entities to negotiate and establish a connection (handshake) before sending any data.	No handshake required; allows entities to send data individually and directly before establishing any set communication path.
Data Format	Uses a continuous byte stream of data.	Uses discrete messages or packets (datagrams).
Packet Pathing	All packets follow the exact same established route (virtual circuit) causing them to arrive perfectly in order.	Each packet carries a destination address and can take different paths; they may arrive out of order.
Robustness	Highly vulnerable to router failure; if the circuit breaks, the connection is instantly lost.	Inherently robust to router failure; datagrams will dynamically find alternative paths.
Protocols	TCP (Transmission Control Protocol), ATM, Frame Relay.	UDP (User Datagram Protocol), IP.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Given entries: 135.46.56.0/22→IF0, 135.46.60.0/22→IF1, 192.53.40.0/23→IF2, default→IF3.

- 135.46.63.10 → matches 135.46.60.0/22 → **Interface 1**
- 192.53.40.7 → matches 192.53.40.0/23 → **Interface 2**
- 135.46.57.14 → matches 135.46.56.0/22 → **Interface 0**
- 192.53.56.7 → no match → **Interface 3 (default)**

Q: ARP (Address Resolution Protocol)

Asked: 2022 Q.6a, 2021 Q.6c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.6a Model Answer
Functionalities of ARP:

- **The Problem:** The network layer routing mechanism utilizes logical 32-bit IP Addresses. However, when a packet arrives at the local network segment, the physical Link Layer hardware only understands 48-bit physical MAC addresses.
- **The Resolution:** Address Resolution Protocol (ARP) precisely bridges this gap. It provides the essential function of dynamically mapping a known logical IP address onto an unknown physical Ethernet (MAC) address.
- **Mechanism (Broadcast Request):** If the sending system does not have the desired MAC address in its local cache, it broadcasts an ARP Request packet onto the entire Ethernet, effectively asking the network, "Who is the owner of this particular IP address?"
- **Mechanism (Unicast Reply):** On receiving that broadcast packet, every machine checks its own configured IP address. Only the correct target machine will respond, sending an ARP Reply containing its physical MAC address directly back to the sender.

(See diagram in the final document: ARP Process)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1, 3 | Slides: Address_Resolution_Protocol_ARP.pdf

Problem: Know the IP, need the MAC. **Solution:** Broadcast ARP Request ("Who has 192.168.1.5?"). Only the owner replies with its MAC via unicast ARP Reply.

Q: NAT (Network Address Translation)

Asked: 2023 Q.5b, 2022 Q.6b, 2019 Q.8b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.5b Model Answer**
> **What NAT Does:**
> Network Address Translation (NAT) allows a single device (like a router) to act as an agent between a local private network and the public Internet. It maps multiple private non-routable IP addresses to a single public IP address, largely solving the IPv4 address exhaustion problem.

Functionality and Translation Process:

- Implementation acts as a proxy: A NAT router maintains an **address translation table**.
- **Outgoing packets:** The NAT router replaces the `(Source IP, Source Port)` of every outgoing datagram from the private host with its own `(NAT Public IP, New Port)`.
- The router remebers this mapping by logging the translation pair in its table.
- **Incoming packets:** When the remote server responds to the `(NAT Public IP, New Port)`, the NAT router looks up that specific port in its translation table, replaces the destination IP/Port in the packet header with the original private `(Source IP, Source Port)`, and forwards it to the correct internal host.

Translation Table Format Example:

Private Connection Side	Public / WAN Side	Remote Target
192.168.1.50:3211	203.0.113.5:60001	104.21.3.4:80
192.168.1.65:5432	203.0.113.5:60002	8.8.8.8:53

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 89, 94 | Slides: Network_Address_Translation_NAT.pdf

Router replaces private (Source IP, Port) with its public (NAT IP, New Port) in a translation table. Replies are looked up and forwarded back to the correct internal host.

Q: DHCP Process

Asked: 2023 Q.8b, 2019 Q.3d

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.8b Model Answer**
> **Why DHCP Process is Needed:**
> Dynamic Host Configuration Protocol (DHCP) automatically provides vital network configuration information (IP address, subnet mask, gateway address) to hosts on lease. Connecting a host manually is highly prone to mistakes (e.g., administrator assigns an already used IP address causing IP conflicts). DHCP dynamically distributes these parameters perfectly, allowing devices to freely log on, move between networks, and request temporary configuration without any manual human intervention.

DHCP DORA Process:

- **D - DHCPDISCOVER:** A new node connects and broadcasts (to `255.255.255.255`) requesting an IP. Source IP is `0.0.0.0`.
- **O - DHCPOFFER:** The DHCP Server picks an available IP and unicassts/broadcasts a proposed offer (IP, lease time) back to the client.
- **R - DHCPREQUEST:** The host selects the offer and officially broadcasts an acceptance to the network to claim it.
- **A - DHCPACK:** The Server officially acknowledges the claim and marks the IP as unavailable in its storage.

(See diagram in the final document: DHCP DORA Process)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 126–127 | Slides: DHCP_Protocol_Breakdown.pdf

DORA Process: (1) **Discover** (client broadcasts). (2) **Offer** (server proposes IP). (3) **Request** (client accepts). (4) **Acknowledge** (server confirms). Lease time limits how long the IP is held.

Q: Tunneling (IPv6 over IPv4)

Asked: 2022 Q.5b

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.5b Model Answer
Differences between IPv4 and IPv6:

Feature	IPv4	IPv6
Address Space	Uses a highly limited 32-bit address space.	Uses a massive 128-bit address space, thoroughly solving IP allocation exhaustion.
Address Hierarchy	Standard network/host hierarchical division.	Features extensively extended address hierarchy allowing complex sub-divisions.
Header Format / Size	Header size varies (typically 20 bytes minimum) due to variable length Options fields.	Features a fixed-size, streamlined base header. The fixed header format drastically helps speed processing up at the router level.
Protocol Field	Contains a standard "Protocol" field to denote the next enclosed payload layer (like TCP).	Replaces Protocol field with a "Next Header" field.
Extension Headers	Options are crammed into the main IPv4 base header, complicating processing.	Options are handled elegantly using distinct modular "Extension headers." Extension headers are placed between the fixed IPv6 base header and the transport level header (TCP/UDP), saving base header space.

[!TIP] **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 97 | Slides: IPv6_Extension_Headers_and_Tunneling.pdf

IPv6 packet is encapsulated inside an IPv4 header to traverse legacy IPv4 networks. The IPv4 "Protocol" field is set to 41 (indicating IPv6 payload).

Q: Why IP is Connectionless; Transparent vs Non-Transparent Fragmentation

Asked: 2021 Q.4c, 2020 Q.6c, 2019 Q.8a

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2021 Q.4c Model Answer
Why IP is Connectionless:
Internet Protocol (Layer 3) is utterly connectionless because it deliberately does not establish, negotiate, or rigidly maintain any formal circuit or path mathematically before sending data. Every single IP packet routed into the internet core acts completely independently. They may explicitly traverse vastly different dynamic physical router paths to get to the same destination, and they might consequently arrive completely out of mathematical order. It strictly leaves complex reliability and strict connection management entirely to the higher Transport layer (TCP).

Transparent vs. Non-Transparent Segmentation (Fragmentation):
When a massive packet definitively hits a network link with a tiny structural MTU boundary, it must be thoroughly broken structurally.

- **Transparent Fragmentation:** An intensely hidden, localized structural approach. A massive packet is shattered tightly by a router upon entering a small-MTU ATM backbone framework. However, the exit-edge router of that specific tiny backbone structurally identically reassembles the fragments completely *before* it gets sent to the rest of the wider internet. The outside world transparently has no idea it was ever internally fragmented.
- **Non-Transparent Fragmentation:** The standard strict IPv4 global approach. When a router forcibly fragments a large packet, the independent fragments structurally remain utterly shattered for the entire remaining geographic journey. Specifically, solely the final ultimate destination host computer aggressively shoulders the burden of definitively reassembling the scattered fragments.

[!TIP] **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 99 | Slides: Transparent_vs_NonTransparent_Fragmentation.pdf

IP is connectionless: No circuit setup before sending. Each packet routed independently. May arrive out of order. Reliability left to TCP.

- **Transparent:** Router fragments at entry to small-MTU network; exit router reassembles before forwarding. Outside world never knows.
- **Non-Transparent (IPv4 standard):** Router fragments; fragments stay split for the entire journey. Only final destination reassembles.

Q: MTU and Path MTU Discovery

Asked: 2023 Q.5c, Q.8c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.5c Model Answer**
> *(Note: Since no specific graph is provided in the general exam template, this outlines the standard Link State algorithmic steps expected for Dijkstra's implementation.)*

Dijkstra's Shortest Path Algorithm (Link State):

1. **Initialization:** The chosen source router (e.g., node A) initializes the cost to itself as 0 and the cost to all other unknown routers as Infinity (∞).
2. **Neighbor Assessment:** The source node examines all its directly connected neighbors. It updates their path costs based on the link state weight (cost) of connecting to them.
3. **Select Minimum:** The node selects the unvisited neighbor with the lowest total path cost from the source and marks it as 'visited'.
4. **Path Relaxation:** From that newly visited node, examine its unvisited neighbors. If the path from the source through this new node to a neighbor is strictly lesser than previously recorded estimated costs, update the routing table with the new, shorter path.
5. **Iteration:** Repeat this greedy selection and relaxation process until all nodes in the network are marked as visited. The result is a shortest-path spanning tree from the source to every other router in the topological database.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: DNS_Lookup_MTU_Discovery.pdf, Network_Season_10_Fragmentation.pdf

MTU: Largest packet size a link can handle (Ethernet = 1500 bytes). **Path MTU Discovery:** Sender sets "Don't Fragment" (DF) bit. If a router can't forward it, it drops the packet and sends ICMP "Fragmentation Needed" with its MTU value. Sender shrinks packet size accordingly.

PART E: ROUTING & CONGESTION CONTROL

Q: Distance Vector vs Link State Routing

Asked: 2023 Q.6a, 2021 Q.5a, 2020 Q.5a, 2019 Q.6a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6a Model Answer**
> **Count to Infinity Problem:**
> Distance Vector routing protocols (like RIP, using the Bellman-Ford algorithm) suffer from the "Count to Infinity" problem due to slow convergence and routing loops.

- **The Issue:** When a link breaks or a network goes completely down, routers take a very long time to realize the path is truly dead. Because routers only share vectors with their immediate neighbors, bad news travels very slowly.
- **The Loop:** If Router X goes offline, Router Y might update its table pointing to Router Z (thinking Z has an alternate path to X). But Z was actually routing to X *through* Y. Y and Z will infinitely cross-update each other, incrementing their "hop count" distance +1 each time, causing packets to bounce in a loop.
- **The Resolution:** The network increments the distance metric indefinitely until it reaches a defined maximum metric (e.g., 16 hops in RIP) which equates to "Infinity", at which point the route is finally dropped. Techniques like "Poison Reverse" and "Split Horizon" are implemented to mitigate this loop.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 81–86 | Slides: Routing_Algorithms_Examples.pdf, Routing_Implementation_Part_1.pdf

- **Distance Vector (Bellman-Ford, e.g., RIP):** Share entire table with neighbors only. Simple, low memory. Slow convergence, vulnerable to routing loops.
- **Link State (Dijkstra, e.g., OSPF):** Flood link status to ALL routers. Each builds a full topology map. Fast convergence, loop-free. Needs more CPU/RAM.

Q: Count-to-Infinity Problem

Asked: 2023 Q.6a, 2021 Q.5d, 2019 Q.6a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6a Model Answer**
> **Count to Infinity Problem:**
> Distance Vector routing protocols (like RIP, using the Bellman-Ford algorithm) suffer from the "Count to Infinity" problem due to slow convergence and routing loops.

- **The Issue:** When a link breaks or a network goes completely down, routers take a very long time to realize the path is truly dead. Because routers only share vectors with their immediate neighbors, bad news travels very slowly.
- **The Loop:** If Router X goes offline, Router Y might update its table pointing to Router Z (thinking Z has an alternate path to X). But Z was actually routing to X *through* Y. Y and Z will infinitely cross-update each other, incrementing their "hop count" distance +1 each time, causing packets to bounce in a loop.
- **The Resolution:** The network increments the distance metric indefinitely until it reaches a defined maximum metric (e.g., 16 hops in RIP) which equates to "Infinity", at which point the route is finally dropped. Techniques like "Poison Reverse" and "Split Horizon" are implemented to mitigate this loop.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 81 | Slides: Routing_and_Congestion_Control.pdf

When a link fails, DV routers keep incrementing hop counts based on stale neighbor info, bouncing packets in loops until the metric reaches "infinity" (16 in RIP).
Solutions: Split Horizon, Route Poisoning.

Q: Default Route, Sink Tree, Route Poisoning, Split Horizon

Asked: 2023 Q.6b, 2022 Q.7b, 2021 Q.6a/b, 2020 Q.5b/c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6b Model Answer**
> | Property | TCP (Transmission Control Protocol) | UDP (User Datagram Protocol) |
> | :--- | :--- | :--- |
> | **Connection Type** | Connection-oriented (requires 3-way handshake) | Connectionless (sends immediately) |
> | **Reliability** | Highly reliable. Ordered delivery guaranteed. | Best-effort delivery. Unreliable. |
> | **Error Checking** | Explicit Acknowledgements (ACK), checksums, retransmission of lost segments. | Only basic header checksum. Missing datagrams are explicitly dropped. |
> | **Flow & Congestion**| Strict Flow control (sliding window) and Congestion control algorithms. | No flow control. No congestion control. |
> | **Header Size** | 20 bytes (minimum) | 8 bytes |
> | **Transmission** | Byte-oriented data stream | Message-oriented independent datagrams |
> | **Use Cases** | File transfer (FTP), Web browsing (HTTP), Email (SMTP, IMAP) | Streaming video, VoIP, DNS, Gaming |

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 78, 81 | Slides: Routing_Questions.pdf

- **Default Route:** Forwarding rule used when no specific route matches. Packets go to default gateway.
- **Sink Tree:** Spanning tree of all shortest paths rooted at the destination.
- **Route Poisoning:** Advertising a dead route with metric=∞ to kill it immediately.
- **Split Horizon:** Never advertise a route back on the interface it was learned from.

Q: Dijkstra's Algorithm (Shortest Path)

Asked: 2023 Q.6b, 2021 Q.6a, 2020 Q.5b, 2019 Q.6b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6b Model Answer**
> | Property | TCP (Transmission Control Protocol) | UDP (User Datagram Protocol) |
> | :--- | :--- | :--- |
> | **Connection Type** | Connection-oriented (requires 3-way handshake) | Connectionless (sends immediately) |
> | **Reliability** | Highly reliable. Ordered delivery guaranteed. | Best-effort delivery. Unreliable. |
> | **Error Checking** | Explicit Acknowledgements (ACK), checksums, retransmission of lost segments. | Only basic header checksum. Missing datagrams are explicitly dropped. |
> | **Flow & Congestion**| Strict Flow control (sliding window) and Congestion control algorithms. | No flow control. No congestion control. |

Header Size	20 bytes (minimum)	8 bytes
Transmission	Byte-oriented data stream	Message-oriented independent datagrams
Use Cases	File transfer (FTP), Web browsing (HTTP), Email (SMTP, IMAP)	Streaming video, VoIP, DNS, Gaming

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 84–86 | Slides: Routing_Algorithms_Examples.pdf

Steps: (1) Set source distance=0, all others=∞. (2) Select unvisited node with lowest cost. (3) Update neighbors if new path is shorter. (4) Mark node visited. (5) Repeat until destination is reached.
(Specific numeric solutions depend on the graph figure provided in each exam paper — see Rifat PDF page 84 for the diagram format.)

Q: Distance Vector Routing Table Calculation

Asked: 2023 Q.5b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.5b Model Answer**
> **What NAT Does:**
> Network Address Translation (NAT) allows a single device (like a router) to act as an agent between a local private network and the public Internet. It maps multiple private non-routable IP addresses to a single public IP address, largely solving the IPv4 address exhaustion problem.

Functionality and Translation Process:

- Implementation acts as a proxy: A NAT router maintains an **address translation table**.
- **Outgoing packets:** The NAT router replaces the `(Source IP, Source Port)` of every outgoing datagram from the private host with its own `(NAT Public IP, New Port)`.
- The router remebers this mapping by logging the translation pair in its table.
- **Incoming packets:** When the remote server responds to the `(NAT Public IP, New Port)`, the NAT router looks up that specific port in its translation table, replaces the destination IP/Port in the packet header with the original private `(Source IP, Source Port)`, and forwards it to the correct internal host.

Translation Table Format Example:

Private Connection Side	Public / WAN Side	Remote Target
`192.168.1.50:3211`	`203.0.113.5:60001`	`104.21.3.4:80`
`192.168.1.65:5432`	`203.0.113.5:60002`	`8.8.8.8:53`

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Problem: Vectors into router C: from B:(5,0,8,12,6,2), from D:(16,12,6,0,9,10), from E:(7,6,3,9,0,4). Link costs: C-B=6, C-D=3, C-E=5.
Method: For each destination, C's cost = min(cost_via_B, cost_via_D, cost_via_E).

- cost_via_X = link_cost(C,X) + X's distance to destination.

Q: Static vs Dynamic Routing; Intra-domain vs Inter-domain

Asked: 2021 Q.5d, 2020 Q.5a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.5d Model Answer**
> **Count-to-Infinity Problem:**
> The fatal mathematical flaw in rudimentary Distance Vector routing. Because routers blindly trust "rumors" passed from immediate neighbors, if a critical link violently drops offline, routers may inadvertently misinterpret obsolete paths as viable alternative routes. They end up pointing at each other in a micro-loop, incrementally increasing the believed path cost by $+1$ helplessly into infinity before mathematically realizing the node is ultimately unreachable.

Static vs. Dynamic Routing:

- **Static Routing:** Network routes are rigidly configured manually by a human administrator. It physically cannot adapt to structural changes or link breaks; if a wire is cut, the network completely fails unless manually reprogrammed. Excellent securely for utterly tiny, isolated, fixed networks.
- **Dynamic Routing:** Routers actively speak specialized languages (RIP/OSPF/BGP) to autonomously negotiate paths. They structurally sense the living topology, share intelligence, and autonomously rebuild routing tables dynamically to logically route entirely around broken hardware links on the fly.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Static:** Manually configured. Cannot adapt to failures. Good for tiny fixed networks.
- **Dynamic:** Routers auto-discover routes (RIP/OSPF/BGP). Adapts to link failures.
- **Intra-domain (IGP):** Routing within one AS (OSPF, RIP). Driven by performance.
- **Inter-domain (EGP):** Routing between AS's (BGP). Driven by policy.

Q: Hierarchical Routing

Asked: 2019 Q.6b

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2019 Q.6b Model Answer**
> **Sink Tree:**
> A sink tree for a destination node is a tree formed by the union of all shortest paths from every other node in the network to that destination. It is rooted at the destination where all traffic converges ("sinks"). Every routing algorithm essentially attempts to discover and use the sink tree for each destination.

(Dijkstra's algorithm step-by-step procedure is described in 2023 Q.6b and 2021 Q.6a. The specific numeric calculation depends on the graph provided in the figure.)

[!TIP] **QUICK REVISION / SHORT NOTES:**

Addresses allocated in contiguous prefix chunks. Routers summarize massive blocks into single prefix routes using CIDR, reducing routing table size dramatically.

Q: Congestion Control — Leaky Bucket, Token Bucket, Choke Packets

Asked: 2023 Q.6c, 2021 Q.5b/c, 2020 Q.6a, 2019 Q.6c

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6c Model Answer**
> **Superiority of Token Bucket:**
> The **Token Bucket algorithm** provides better overall network performance and flexibility compared to the Leaky Bucket algorithm.

Why (Based on Implementation Differences):

- **Handling Burstiness:** An important difference is that Leaky Bucket enforces a very rigid pattern on the output stream, discarding packets immediately when the bucket is full. It forces bursty traffic to smooth out entirely into a fixed, constant rate.
- **Permitting Capacity:** Token bucket is less restrictive. It permits burstiness but bounds it. It collects tokens generated at a regular rate. Packets waiting in a queue must grab a token before being transmitted. If a sudden burst of data arrives and the bucket has saved up tokens, those bursts of packets can be sent all at once.
- **Discard Policy:** Token bucket throws away excess *tokens* when the bucket is full, but never drops the actual user *packets* (they simply wait in queue until new tokens generate). Leaky bucket drops *packets* when full. Thus, token bucket adapts far better to the bursty nature of modern web traffic without causing data loss.

[!TIP] **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 88–93, 142 | Slides: QoS_Leaky_Token_Bucket_Loadshedding.pdf, Token_Bucket_vs_Leaky_Bucket.pdf

- **Leaky Bucket:** Converts bursty traffic to fixed-rate output. Overflows = packets dropped. Strict policing.
- **Token Bucket:** Tokens added at fixed rate. Must spend token to send. Allows burst (up to bucket size). Drops excess *tokens*, not packets. **Better performance than Leaky Bucket.**
- **Hop-by-Hop Choke Packet:** Congested router sends choke signal to EVERY intermediate hop on the backward path, not just source. Provides immediate relief.

Q: Load Shedding and QoS

Asked: 2021 Q.6b

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.6b Model Answer**
> **(i) Route Poisoning & Split Horizon:**

- **Route Poisoning:** Immediately broadcasting a freshly broken link forcefully with a strict metric of "Infinity" (e.g., metric 16 in RIP) to immediately kill the route globally, overriding all other rumors and preventing Count-to-Infinity.
- **Split Horizon:** A mathematical rule barring a router from illegally broadcasting a learned routing metric right back exactly out the physical interface it originally learned it from, killing simple two-node ping-pong routing loops instantly.

(ii) Load Shedding & QoS:

- **Load Shedding:** The "big hammer" approach. When intense congestion forces a router to overflow, it just starts throwing out packets blindly to survive. Simplest way is to drop packets randomly, but discard policy may depend on the application.
- **QoS (Quality of Service):** Replaces blind dropping. QoS implements extremely strict parameterization algorithms allowing critical time-sensitive traffic explicit priority handling in router queues (e.g., header changes to facilitate QoS).

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Load_Shedding_Wine_vs_Milk_Packets.pdf, Introduction_to_QoS.pdf

- **Load Shedding:** Router drops packets when overwhelmed. May prioritize (e.g., keep newer "milk" packets over old "wine" packets).
- **QoS:** Strict priority queuing for time-sensitive traffic. Uses DiffServ/IntServ.

Q: Flow Control vs Congestion Control

Asked: 2023 Q.7a, 2022 Q.7a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.7a Model Answer**
>
> - **Flow Control:** This is a vital mechanism regulating the transmission of data strictly between two end devices (Sender and Receiver). It considers only what is going on within that specific session. Its primary goal is to ensure a fast sender does not overwhelm a slow receiver. It is managed by advertising a receive window size.
> - **Congestion Control:** This operates globally. It regulates the flow of packets coming into the network as a whole, preventing the entire network fabric (routers and links) from becoming overloaded. Its goal is to prevent network saturation, dropped packets, and transmission delays across all users.
> - *Interaction:* In TCP, the Effective Window Size (the actual amount of data the source can send) is dictated by taking the mathematically stricter of the two: `MIN(CongestionWindow, AdvertisedWindow)`.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Flow Control:** Point-to-point. Prevents fast sender from overwhelming slow receiver (advertised window).
- **Congestion Control:** Global. Prevents network saturation (congestion window). TCP effective window = min(cwnd, advertised window).

Q: Datagram vs Virtual Circuit

Asked: 2023 Q.5a, 2022 Q.7b

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.5a Model Answer**
> > **Connection-oriented Service:**
>
> - **Definition:** Analogous to the telephone system. It requires communication entities to use a handshake process to establish a connection before sending data. TCP provides connection-oriented services.
> - **Properties:** Transmits data as a continuous stream of bytes. Reliability is strictly achieved by having the recipient acknowledge each message. Packets are received in order due to sequencing and flow control. Uses virtual circuit switching. It is vulnerable to router failure along the dedicated path.
> - **Application Examples:** File transfer (FTP), remote login (Telnet, SSH), and video on demand. These require guaranteed, ordered delivery.
>
> **Connection-less Service:**
>
> - **Definition:** Analogous to the postal system. Packets of data (datagrams) are transmitted from source to destination directly without establishing a connection. UDP and IP provide connectionless service.
> - **Properties:** Handles packets individually. Each packet carries a full destination address. Packets do not follow a fixed path and can arrive out of order. It is robust to router failure since alternate paths can be taken dynamically.
> - **Application Examples:** Credit card point-of-sale verification, electronic funds transfer, remote database access queries, DNS. These are inherently query/reply where speed and robustness are preferred over sustained stream overhead.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 75, 90–91 | Slides: Datagram_vs_Virtual_Circuit_Packet_Switching.pdf

- **Virtual Circuit:** Path established before data flows. Packets follow same route, arrive in order. Vulnerable to link failure.
- **Datagram:** Each packet independent, different paths possible. Robust to failure. May arrive out of order.
- **For low-latency trading:** Virtual Circuit recommended for consistent pathing.

PART F: APPLICATION LAYER & SECURITY

Q: DNS — Namespace and Resolver Lookup

Asked: 2021 Q.7a/b, 2020 Q.7a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.7a Model Answer**
> **DNS Namespace:**
> The DNS namespace physically strictly resembles a massive inverted tree. The absolute root is the Root Domain (a simple dot `.`). Branched below are Top-Level Domains (TLDs like `.com`, `.edu`, `.bd`). Below them are Second-Level Domains (`google.com`, `cuet.ac.bd`). This decentralized hierarchy prevents catastrophic naming collisions completely across the entire planet.

Resolver Steps (Resolving `cuet.ac.bd` from MIT):

1. **Local Check:** User types URL. PC asks strictly its Local MIT ISP DNS Cache. If missing, proceed.
2. **Root Server Query:** Local DNS formally asks the Internet Root Server (`.`). The Root server mathematically knows nothing of CUET, but authoritatively returns the IP address belonging uniquely to the `.bd` TLD Server.
3. **TLD Server Query:** Local DNS asks the `.bd` TLD Server. It uniquely returns the structural IP belonging to the `.ac.bd` generic registry.
4. **Authoritative Server Query:** Local DNS definitively asks the `.ac.bd` registry. The registry explicitly returns the final, authoritative hardware IP explicitly mapped rigidly to `cuet.ac.bd`'s physical server.
5. **Caching & Delivery:** Local MIT DNS physically caches the result to accelerate the next immediate request globally and firmly returns it to the host PC application.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 123, 126–127 | Slides: Application_Layer_Domain_Name_System.pdf, DNS_Hierarchical_Architecture.pdf

DNS Namespace: Inverted tree. Root → TLDs (.com, .bd) → Second-level (cuet.ac.bd).
Resolver Steps (MIT resolving cuet.ac.bd):

1. Check local cache → 2. Query Root Server → returns .bd TLD server IP → 3. Query .bd → returns .ac.bd IP → 4. Query .ac.bd authoritative server → returns cuet.ac.bd IP → 5. Cache and return.

Q: Short Notes — SSH, FTP, SMTP, HTTP/HTTPS, WWW

Asked: 2022 Q.8c, 2021 Q.7c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.8c Model Answer**
> **Differences between EGP and OSPF:**

Feature	Exterior Gateway Protocol (EGP / BGP)	Open Shortest Path First (OSPF)
Routing Scope	Categorized as an Exterior Gateway Protocol.	Categorized as an Interior Gateway Protocol (IGP).
Operational Boundary	Specifically designed to route packets logically *between* entirely independent Autonomous Systems (AS's) on the global internet.	Specifically designed to route packets strictly *within* a single, localized Autonomous System using common internal metrics.
Underlying Algorithm	Usually based on generic Distance-Vector logic (Path Vector).	Uses complex Link-State routing algorithms and metrics based strictly on link costs.
Supernetting Support	Older EGP protocols do not support supernetting / CIDR aggregation directly, though its modern successor BGP does perfectly.	Fully supports CIDR and Supernetting out of the box, drastically simplifying massive internal routing tables.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 4, 112 | Slides: Application_Layer_SMTP_and_FTP.pdf, HTTP_Fundamentals.pdf

- **SSH (Port 22):** Encrypted remote terminal access, replaces Telnet.
- **FTP (Port 20/21):** Bulk file transfer between client/server.
- **SMTP (Port 25):** Push protocol for sending outgoing email.
- **HTTP (Port 80):** Web page request/response. **HTTPS (Port 443):** HTTP + TLS/SSL encryption.
- **WWW:** Global system of interlinked hypertext documents accessed via URLs.

Q: SMTP, POP3, IMAP — Email Protocols

Asked: 2023 Q.8a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.8a Model Answer**
> **Transposition Cipher Mathematics:**
> A transposition cipher does not substitute characters; rather, it performs a permutation (reordering) on the plaintext characters.

- **Key:** In a transposition cipher, the encryption key is explicitly the mathematical rule defining the permutation of character positions. Let the message length be L. A permutation pattern is defined over a block size b. Let permutation function be $f(i)$.
- **Encryption:** The cipher function takes the plaintext character at position i and moves it to a new position $j = f(i)$.
  $C[f(i)] = P[i]$
- **Decryption:** The receiver uses the inverse mathematical permutation using the same block length.
  $P[i] = C[f^{-1}(i)]$
- Example: In a Rail Fence or columnar transposition, the exact geometry of rows/columns dictates the mathematical permutation array utilized.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Electronic_Mail_SMTP_POP_IMAP.pdf, How_Email_Works_SMTP_POP3_IMAP.pdf

- **SMTP (Port 25):** Sending/relaying mail. Push only.
- **POP3 (Port 110):** Downloads mail, deletes from server. Bad for multi-device use.
- **IMAP (Port 143):** Syncs mail across all devices. Stays on server.

Q: ACL — Standard vs Extended

Asked: 2023 Q.7a, 2022 Q.8a, 2021 Q.6c/d, 2020 Q.5c/d, 2019 Q.5c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7a Model Answer**

- **Flow Control:** This is a vital mechanism regulating the transmission of data strictly between two end devices (Sender and Receiver). It considers only what is going on within that specific session. Its primary goal is to ensure a fast sender does not overwhelm a slow receiver. It is managed by advertising a receive window size.
- **Congestion Control:** This operates globally. It regulates the flow of packets coming into the network as a whole, preventing the entire network fabric (routers and links) from becoming overloaded. Its goal is to prevent network saturation, dropped packets, and transmission delays across all users.
- *Interaction:* In TCP, the Effective Window Size (the actual amount of data the source can send) is dictated by taking the mathematically stricter of the two: `MIN(CongestionWindow, AdvertisedWindow)`.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 134–137 | Slides: Firewalls_and_Virtual_Private_Networks.pdf

- **Standard ACL (1-99):** Filters by Source IP only. Place close to **destination**.
- **Extended ACL (100-199):** Filters by Source, Destination, Protocol, Port. Place close to **source** (saves bandwidth).

Q: Cryptography — Substitution & Transposition Ciphers

Asked: 2023 Q.7b, 2019 Q.7a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7b Model Answer**
> **Virtual Circuit Packet Switching:**

- Operates like circuit switching dynamically. Before data flows, a virtual path is established edge-to-edge.
- Each packet contains a Virtual Circuit Identifier (VCI) in its header.
- **Routing Decision:** No complex routing decisions are required at each node; routers simply look up the VCI and forward along the pre-established path.
- Provides Quality of Service (QoS), guaranteed sequential delivery, and minimal per-packet processing delay, but is highly vulnerable to link failure (the whole circuit drops).

Datagram Packet Switching:

- Operates purely connectionless. Every packet carries full source and destination IP addresses.
- **Routing Decision:** Routing decisions are deeply made dynamically for each individual datagram at every single router hop.
- Datagrams can follow entirely different routes to the destination and arrive out of order.
- Highly robust. If a router malfunctions, datagrams simply route around the failure dynamically.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 138 | Slides: Network_Security_Module_8_Part_2.pdf

- **Substitution (Caesar):** Replace each letter (e.g., shift 3: A→D). Vulnerable to frequency analysis.
- **Transposition:** Rearrange letter order using a keyword matrix. Letters unchanged, positions scrambled.

Q: Vigenère Cipher vs Caesar Cipher

Asked: 2022 Q.6c
Slides: Network_Security_Module_8_Part_2.pdf

- **Caesar:** Mono-alphabetic, uniform shift. Easily broken by frequency analysis.
- **Vigenère:** Poly-alphabetic, variable shift per letter using a keyword. Flattens frequency distribution, making analysis much harder. **Yes, Vigenère is better.**

Q: DES Algorithm Principles

Asked: 2023 Q.7c, 2022 Q.8b, 2019 Q.7b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7c Model Answer**
> The Internet protocol suite (commonly known as TCP/IP or DoD model) consists of four abstraction layers:

1. **Application Layer:** Contains all protocols for specific data communications services on a process-to-process level. Ensures network services are available to applications (HTTP, FTP, SMTP, DNS).
2. **Transport Layer:** Handles host-to-host communication. It establishes logical end-to-end connections managing flow control, reliability, and segmentation (TCP, UDP).
3. **Internet Layer (Network):** Handles connecting independent networks and establishing internetworking. Specifies how data should be formatted, addressed, and routed to the destination (IP, ICMP, ARP).
4. **Network Access (Link Layer):** Contains communication technologies for a single physical network segment. It combines OSI Data Link and Physical layers, handling framing, MAC addressing, and hardware interfacing (Ethernet, Wi-Fi, PPP).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 129 | Slides: RSA_Cryptography.pdf

DES: Symmetric block cipher. 64-bit blocks, 56-bit key, 16-round Feistel structure. Each round: expansion, XOR with round key, S-box substitution, P-box permutation. Now insecure due to short key; replaced by AES.

Q: RSA Algorithm — Encrypt "COMPUTER"

Asked: 2023 Q.7c, 2021 Q.8a/b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7c Model Answer**
> The Internet protocol suite (commonly known as TCP/IP or DoD model) consists of four abstraction layers:

1. **Application Layer:** Contains all protocols for specific data communications services on a process-to-process level. Ensures network services are available to applications (HTTP, FTP, SMTP, DNS).
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3. **Internet Layer (Network):** Handles connecting independent networks and establishing internetworking. Specifies how data should be formatted, addressed, and routed to the destination (IP, ICMP, ARP).
4. **Network Access (Link Layer):** Contains communication technologies for a single physical network segment. It combines OSI Data Link and Physical layers, handling framing, MAC addressing, and hardware interfacing (Ethernet, Wi-Fi, PPP).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 129 | Slides: RSA_Cryptography.pdf

RSA Steps: (1) Choose primes p, q. (2) $n = p \times q$. (3) $\varphi(n)=(p-1)(q-1)$. (4) Choose e coprime to $\varphi(n)$. (5) $d = e^{-1} \bmod \varphi(n)$. **Encrypt:** $C = M^e \bmod n$. **Decrypt:** $M = C^d \bmod n$. Convert each letter of "COMPUTER" to its numeric value, apply encryption formula.

Q: Transposition Cipher Problems (Solved)

Asked: 2023 Q.7c (QUESTION), 2021 Q.8c (DICTIONARY), 2020 Q.8b (LOCKDOWN), 2019 Q.7c (QUESTION), 2019 Q.8b (MEGABUCK)

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7c Model Answer**
> The Internet protocol suite (commonly known as TCP/IP or DoD model) consists of four abstraction layers:

1. **Application Layer:** Contains all protocols for specific data communications services on a process-to-process level. Ensures network services are available to applications (HTTP, FTP, SMTP, DNS).
2. **Transport Layer:** Handles host-to-host communication. It establishes logical end-to-end connections managing flow control, reliability, and segmentation (TCP, UDP).
3. **Internet Layer (Network):** Handles connecting independent networks and establishing internetworking. Specifies how data should be formatted, addressed, and routed to the destination (IP, ICMP, ARP).
4. **Network Access (Link Layer):** Contains communication technologies for a single physical network segment. It combines OSI Data Link and Physical layers, handling framing, MAC addressing, and hardware interfacing (Ethernet, Wi-Fi, PPP).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Key="DICTIONARY", Plain="happinessistemporary"

Alphabetical order: A(8), C(3), D(1), I_1(2), I_2(5), N(7), O(6), R(9), T(4), Y(10)
Cipher: `SAPEHSATIPERNOSRPMIY`

Key="LOCKDOWN", Plain="stayhomestaysafe"

Alphabetical: C(3), D(5), K(4), L(1), N(8), O_1(2), O_2(6), W(7)
Cipher: `AAHSYYSSEETTOAMF`

Key="QUESTION", Plain="Weloveourcountrybangladesh"

Alphabetical: E(3), I(6), N(8), O(7), Q(1), S(4), T(5), U(2)
Cipher: `LONXETAXUYEXORDXWRBSOUGXVNLXECAH`

Key="MEGABUCK", Plain="pleasetransferfivelatektoinmysonalibankaccount"

Alphabetical: A(4), B(5), C(7), E(2), G(3), K(8), M(1), U(6)
Cipher: `AFAMBUSETYANTFKOKXLNEILCESLNIORITNAXPAVOACERESNT`

Q: Short Notes — ICMP, NAT, Digital Signature, FDDI, Firewall, VPN, VLAN

Asked: 2022 Q.8c, 2021 Q.7c, 2019 Q.8b/c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.8c Model Answer**
> **Differences between EGP and OSPF:**

Feature	Exterior Gateway Protocol (EGP / BGP)	Open Shortest Path First (OSPF)
Routing Scope	Categorized as an Exterior Gateway Protocol.	Categorized as an Interior Gateway Protocol (IGP).
Operational Boundary	Specifically designed to route packets logically *between* entirely independent Autonomous Systems (AS's) on the global internet.	Specifically designed to route packets strictly *within* a single, localized Autonomous System using common internal metrics.
Underlying Algorithm	Usually based on generic Distance-Vector logic (Path Vector).	Uses complex Link-State routing algorithms and metrics based strictly on link costs.
Supernetting Support	Older EGP protocols do not support supernetting / CIDR aggregation directly, though its modern successor BGP does perfectly.	Fully supports CIDR and Supernetting out of the box, drastically simplifying massive internal routing tables.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 70–73 (VLAN/FDDI), 89 (NAT), 140 (VPN) | Slides: Various

- **ICMP:** Error/diagnostic messages alongside IP (Ping, Traceroute, "Destination Unreachable").
- **Digital Signature:** Sender encrypts document hash with Private Key. Receiver verifies with sender's Public Key. Ensures non-repudiation.
- **Firewall:** Stateful security device between internal network and internet. Implements ACLs + deep packet inspection.
- **VPN:** Encrypted tunnel over public network for secure remote access.
- **VLAN:** Logical segmentation of a physical switch into multiple virtual subnets.

PART G: MODERN TOPICS (2023 New — Not in Rifat PDF)

Q: IoT, Cloud Computing, Autonomous Vehicles — Client-Server vs P2P

Asked: 2023 Q.4a, 2022 Q.1a

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.4a Model Answer**
> **IP Address Classes and Ranges:**
> Traditional IPv4 addresses are divided into classes to accommodate networks of varying sizes.

- **Class A:** `0.0.0.0` to `127.255.255.255`. Designed for extremely large networks.
- **Class B:** `128.0.0.0` to `191.255.255.255`. Designed for medium-sized networks.
- **Class C:** `192.0.0.0` to `223.255.255.255`. Designed for smaller local area networks.
- **Class D:** `224.0.0.0` to `239.255.255.255`. Reserved strictly for Multicast addressing.
- **Class E:** `240.0.0.0` to `255.255.255.255`. Reserved for experimental/future use.

Components to Determine Identifiers:
An IP address is logically broken into two primary components, separated using a subnet mask or CIDR notation:

1. **Network Prefix (Network ID):** A common group of most-significant bits used for routing datagrams to the correct specific network.
2. **Rest Field (Host ID):** The remaining bits used to uniquely identify a specific host/node within that localized network.

[!TIP] **QUICK REVISION / SHORT NOTES:**

- **Cloud/IoT:** Client-Server model. Sensors/devices push data to central cloud servers.
- **Autonomous Vehicles:** Hybrid. Client-Server for map/traffic updates + P2P (Vehicle-to-Vehicle) for instant braking alerts.

Q: 5G Technology Features

Asked: 2023 Q.7a

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7a Model Answer**

- **Flow Control:** This is a vital mechanism regulating the transmission of data strictly between two end devices (Sender and Receiver). It considers only what is going on within that specific session. Its primary goal is to ensure a fast sender does not overwhelm a slow receiver. It is managed by advertising a receive window size.
- **Congestion Control:** This operates globally. It regulates the flow of packets coming into the network as a whole, preventing the entire network fabric (routers and links) from becoming overloaded. Its goal is to prevent network saturation, dropped packets, and transmission delays across all users.
- *Interaction:* In TCP, the Effective Window Size (the actual amount of data the source can send) is dictated by taking the mathematically stricter of the two: `MIN(CongestionWindow, AdvertisedWindow)`.

[!TIP] **QUICK REVISION / SHORT NOTES:**

(1) Up to 10 Gbps bandwidth. (2) ~1ms latency (URLLC). (3) 1M+ devices/km² (mMTC). (4) Network Slicing.

Q: Access Networks

Asked: 2023 Q.1b

[!NOTE] **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.1b Model Answer**
> **Advantages of UTP over STP:**

- **Cost-Effectiveness:** UTP is significantly less expensive to manufacture and purchase than STP because it lacks the internal foil or braided shielding.
- **Installation & Flexibility:** UTP has a much smaller outer diameter and is more flexible, making it easier to install, route through tight conduits, and terminate than the bulkier STP cable.
- **Maintenance:** UTP requires less maintenance. STP shielding must be properly grounded at both ends; if grounding is imperfect, the shield acts as an antenna, drawing in more noise than it blocks. UTP eliminates this complex grounding requirement.
- **Adequate for Most Usage:** For most indoor, standard LAN topologies (Voice grade Cat 3, Data grade Cat 5 up to 100 Mbps), UTP provides perfectly adequate transmission without the added bulk and cost of STP, especially in Star topologies.

[!TIP] **QUICK REVISION / SHORT NOTES:**

The "last mile" connecting homes to ISP. Critical for bandwidth (symmetrical for video calls), latency (real-time), and scalability (massive simultaneous users).

Q: Cloud vs Quantum Computing

Asked: 2023 Q.3c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.3c Model Answer**
> **Technique Specified:**
> This specifies the **Cyclic Redundancy Check (CRC)** error detection technique, utilizing polynomial arithmetic done modulo 2 (which is effectively XOR arithmetic without carries).

Given:

- Message (`M`): `1101`
- Message appended with 3 zeros for division (`k` as per lecture): `1101000`
- Generator Polynomial / Divisor (`P`): $x^3 + x + 1$ which translates to binary `1011`. (Since we appended 3 zeros, the divisor must be 4 bits).

Calculation (Modulo-2 Division):
We divide `1101000` by `1011` using XOR:

```
         1110  (Quotient)
     ________
1011 | 1101000
       1011
       ----
        1100
        1011
        ----
         1110
         1011
         ----
          1010
          0000
          ----
           101  (Remainder) ---> Actually, let's recalculate precisely:
```

Precise Modulo-2 XOR Steps:

1. `1101` XOR `1011` = `0110`
2. Bring down `0` -> `1100`
3. `1100` XOR `1011` = `0111`
4. Bring down `0` -> `1110`
5. `1110` XOR `1011` = `0101`
6. Bring down `0` -> `1010`
7. `1010` XOR `1011` = `0001`

- **Remainder (r):** `001`

Final Data Sent:
The CRC remainder `001` is appended to the original message `1101`.
The bit frame `(k+r)` actually transmitted is **`1101001`**.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Cloud:** Classical bits (0 or 1). General-purpose, scalable via internet.
- **Quantum:** Qubits (superposition: 0, 1, or both). Exponentially faster for specific problems (e.g., breaking RSA).

Q: Cyber-Attacks Overview

Asked: 2023 Q.7b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.7b Model Answer**
> **Virtual Circuit Packet Switching:**

- Operates like circuit switching dynamically. Before data flows, a virtual path is established edge-to-edge.
- Each packet contains a Virtual Circuit Identifier (VCI) in its header.
- **Routing Decision:** No complex routing decisions are required at each node; routers simply look up the VCI and forward along the pre-established path.
- Provides Quality of Service (QoS), guaranteed sequential delivery, and minimal per-packet processing delay, but is highly vulnerable to link failure (the whole circuit drops).

Datagram Packet Switching:

- Operates purely connectionless. Every packet carries full source and destination IP addresses.
- **Routing Decision:** Routing decisions are deeply made dynamically for each individual datagram at every single router hop.
- Datagrams can follow entirely different routes to the destination and arrive out of order.
- Highly robust. If a router malfunctions, datagrams simply route around the failure dynamically.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

(1) **DDoS:** Flood server with junk traffic. (2) **Phishing:** Trick humans into revealing passwords. (3) **MITM:** Intercept communication between two parties. (4) **Ransomware:** Encrypt victim's data, demand payment.

Q: Miscellaneous Comparisons

Asked across multiple years

Comparison	Key Difference
Hub vs Switch	Hub broadcasts; Switch forwards to specific port
Bridge vs Router	Bridge = L2 (MAC); Router = L3 (IP)
Internet vs Intranet	Public global vs. Private internal
Loopback vs Physical Address	127.0.0.1 (software test) vs MAC (hardware burned-in)
Message Switching vs Packet Switching	Whole message stored-and-forwarded vs. broken into packets pipelined
Connection-oriented vs Connectionless	TCP (handshake, reliable) vs UDP/IP (no setup, best-effort)
TP (Twisted Pair) vs Fiber	Copper, shorter distance vs. Glass, longer distance, higher speed
Data Link vs Network Layer	MAC/Framing (L2) vs. IP/Routing (L3)

PART H: ADDITIONAL QUESTIONS FOUND IN GAP ANALYSIS

> These questions were identified as missing during the cross-reference audit.

Q: What is Ethernet? What is Gigabit Ethernet?

Asked: 2021 Q.1b, 2019 Q.1c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.1b Model Answer**
> **Ethernet:**
> Ethernet (standardized precisely as IEEE 802.3) is the overwhelmingly dominant ubiquitous technology for Local Area Networks (LANs). It fundamentally defines the wiring, signaling standards, and MAC addressing logic physically at Layer 1 (Physical) and broadly at Layer 2 (Data Link). It traditionally utilizes CSMA/CD for collision management over twisted pair cables, though modern switched Ethernet practically eliminates collisions entirely.

Different Types of Network Topologies:
(Discussed fully previously. Essential forms include:)

- **Bus:** Single shared central coaxial backbone. Prone to cascading failure.
- **Star:** End nodes branching from a central intelligent switch. Modern Ethernet standard.
- **Ring:** Circular node-to-node token passing. Obsolete.
- **Mesh:** Interconnected web of nodes providing supreme fault-tolerant redundancy.
- **Tree/Hybrid:** Branching Star topologies connected via a core Bus spine.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 61–62 | Slides: IEEE_802_3_Ethernet_Standard.pdf, Ethernet_CSMACD_Exponential_Backoff.pdf

- **Ethernet (IEEE 802.3):** The dominant LAN technology. Defines wiring, signaling, and MAC addressing at Layers 1-2. Originally used CSMA/CD over coaxial bus; modern switched Ethernet eliminates collisions.
- **Gigabit Ethernet (IEEE 802.3z/ab):** Operates at 1000 Mbps (1 Gbps). Uses full-duplex point-to-point switched links. CSMA/CD is technically defined but practically not needed in full-duplex mode. Carrier Extension extends minimum frame size to 512 bytes for half-duplex Gigabit when needed.

Q: Which OSI Layer Handles What?

Asked: 2021 Q.2b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.2b Model Answer**
> i) **To route packets:** Network Layer (Layer 3)
> ii) **To run services like FTP and Telnet:** Application Layer (Layer 7)
> iii) **To detect and correct error:** Data Link Layer (Layer 2) primarily; or optionally Transport Layer (Layer 4) for end-to-end.
> iv) **To convert packets to frame:** Data Link Layer (Layer 2)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1–4

Task	OSI Layer
Routing packets	Network Layer (Layer 3)
Running FTP and Telnet services	Application Layer (Layer 7)
Detecting and correcting errors	Data Link Layer (Layer 2)
Converting packets to frames	Data Link Layer (Layer 2)

Q: 5-Layer Protocol Hierarchy — Message Delivery Through Switch and Router

Asked: 2022 Q.2a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.2a Model Answer**
> **Definitions:**

- **Physical Topology:** Refers to the physical layout and actual geometric arrangement of nodes, cables, and switches in a network.
- **Logical Topology:** Refers to how data actually flows and is transmitted through the network from one device to the next, regardless of the physical layout (e.g., passing a token logically in a ring over a physical star wiring).

Advantages and Disadvantages of Topologies:

- **Bus Topology:**
  - **Advantages:** Simple layout. Easy to connect a computer or peripheral to a linear bus. Requires less cable length than a star topology.
  - **Disadvantages:** Entire network shuts down if there is a break in the main backbone cable. Terminators are required at both ends. Difficult to identify the problem if the entire network goes down.
- **Ring Topology:**
  - **Advantages:** Every node gets equal access to the token (predictable delay). Performs better than a bus topology under a heavy network load.
  - **Disadvantages:** A break in the closed ring cable or the failure of a single node can bring down the entire network (unless special dual-ring bypass hardware is used). Difficult to add or remove nodes without disrupting the active ring.
- **Star Topology:**
  - **Advantages:** High speed data transfer; particularly when the star coupler acts as a switch. Easiest to maintain and troubleshoot. Very flexible. A failure of one node or link does not affect the rest of the network.
  - **Disadvantages:** Complete dependency on the central node/coupler; if it fails, the entire network drops. Cabling costs are higher because the number of links is strictly proportional to n (every node needs an independent cable).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 2–4 | Slides: Introduction_to_Layering.pdf

A message of length m bytes at Application layer gets headers added at each layer:

```
Application:    [AH | m bytes]                  → m + h_a bytes
Transport:      [TH | AH | m]                   → m + h_a + h_t bytes
Network:        [NH | TH | AH | m]              → m + h_a + h_t + h_n bytes
Data Link:      [DH | NH | TH | AH | m | DT]    → + h_d + trailer
Physical:       Raw bits on wire

At a Link Layer Switch (Layer 2): Strips DH/DT → reads MAC → re-adds new DH/DT → forwards
At a Router (Layer 3): Strips DH/DT → reads NH (IP) → makes routing decision → adds new DH/DT → forwards
```

Q: Reasons for Using Layered Protocols

Asked: 2019 Q.2a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2019 Q.2a Model Answer**
> > **Reasons for Layered Protocols:**
>
> 1. **Modularity (Divide & Conquer):** It intimately breaks down the immense, monstrous complexity of global network communications firmly into much smaller, actively manageable, and well-defined mathematical sub-tasks.
> 2. **Interoperability:** It strictly standardizes heavily the interfaces globally between completely different vendors, eagerly allowing hardware from Cisco to seamlessly fundamentally talk securely to hardware from Juniper.
> 3. **Flexibility & Upgradeability:** A protocol completely within one specific layer can be fully replaced or modernized securely (e.g., eagerly upgrading from IPv4 to IPv6 at Layer 3) structurally without forcing any changes universally to the overlying applications (Layer 7) or underlying physical cables (Layer 1).
> 4. **Troubleshooting:** Facilitates remarkably easier network fault isolation structurally by allowing engineers actively to strictly thoroughly test layers distinctly one step at a time explicitly from bottom-up or top-down.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1–2

1. **Modularity:** Each layer can be designed, tested, and updated independently.
2. **Abstraction:** Upper layers don't need to know the details of lower layers' implementation.
3. **Interoperability:** Standardized interfaces between layers allow equipment from different vendors to communicate.
4. **Ease of Maintenance:** A bug in one layer can be fixed without redesigning the entire protocol stack.
5. **Reusability:** Lower layers (Physical, Data Link) can be reused across many different applications.

Q: Hidden Station Problem

Asked: 2022 Q.3b

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2022 Q.3b Model Answer**
> > **Explanation of the Term:**
> > In connectionless network systems like the Internet Protocol (IP), communication is analogous to the postal system. Each packet of data (datagram) is treated as an individual, independent entity traversing the network.
>
> - **The Identifier:** To ensure successful delivery, each datagram header strictly carries an "IP Address" functioning as the precise destination address to identify the intended recipient host.
> - **Format:** The IPv4 address is a 32-bit (4-byte) numerical identifier, typically represented in dotted-decimal format (e.g., `192.168.1.10`).
> - **Addressing Datagrams:** Routers across the internet rely completely on this destination address embedded within the packet header. The routing algorithm decides where the packet goes next (the output line) solely by examining this 32-bit identifier against its routing tables.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Wireless_MAC_Protocols.pdf

Problem: In wireless networks, Station A and Station C can both communicate with Station B but are out of range of each other. A cannot "hear" C transmitting. Both may transmit to B simultaneously, causing a collision at B that neither A nor C can detect.

Solution: CSMA/CA with **RTS/CTS** handshake. Before transmitting, A sends a Request-To-Send (RTS) to B. B replies with Clear-To-Send (CTS) which is heard by ALL stations in B's range (including C). C now knows B is reserved and stays silent until A's transmission concludes.

Q: IPv6 Header Format

Asked: 2022 Q.6b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.6b Model Answer**
> **What is NAT?**
> Network Address Translation (NAT) is a technique deployed on routers acting as an intermediary between a private internal network and the public Internet. It allows tens of thousands of simultaneous internal connections to share a single, routable public LAN-side WAN IP address by dynamically replacing IPs and ports.

The Controversy:
NAT is considered "controversial" in traditional open networking because it fundamentally violates the pure end-to-end architectural principle of the Internet. According to strict TCP/IP design, every endpoint should have a unique, globally routable IP and address other endpoints directly. NAT obfuscates the true endpoints, complicating peer-to-peer protocols (like certain VoIP setups or direct file transfers), which struggle to traverse NAT firewalls since the public IP does not truly belong to the internal endpoint.

Functionality and Translation Table:

1. **Outgoing:** When an internal host (e.g. `H1`) sends a datagram to the Internet, the NAT router intercepts it. It explicitly replaces the `(source IP address, original port #)` of the outgoing datagram with its own `(NAT Public IP address, newly assigned port #)`.
2. **Tracking:** The router rigorously logs this translation pair in its internal **NAT translation table** mapping.
3. **Incoming:** When the remote server logically responds to the `(NAT Public IP address, new port #)`, the NAT device intercepts the reply, looks up the port in its translation table, restores the original `(source IP address, original port #)`, and forwards it inwards to `H1`.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 102 | Slides: IPv6_Extension_Headers_and_Tunneling.pdf

Fixed 40-byte IPv6 base header fields:

```
| Version (4b) | Traffic Class (8b) | Flow Label (20b)    |
| Payload Length (16b)  | Next Header (8b) | Hop Limit (8b)|
|             Source Address (128 bits)                    |
|           Destination Address (128 bits)                 |
```

- **Version:** Always 6. **Traffic Class:** QoS priority. **Flow Label:** Identifies a flow for special handling.
- **Next Header:** Replaces IPv4's "Protocol" field. Points to extension headers or upper layer (TCP=6, UDP=17).
- **Hop Limit:** Replaces TTL. Decremented by 1 at each router; packet discarded at 0.

Q: LAN vs WAN Differences

Asked: 2022 Q.2b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.2b Model Answer**
> | Feature | LAN (Local Area Network) | WAN (Wide Area Network) |
> | :--- | :--- | :--- |
> | **Geographical Area** | Restricted to a limited geographical coverage such as a house, single office building, or school campus. | Spans a large geographic area, such as a state, province, or country. |
> | **Data Transmission Cost** | Much lower, since the data transmission medium is usually owned outright by the user organization. | Very high, because the transmission mediums used are leased lines or public carrier systems (telephone lines, microwaves, satellites). |
> | **Physical Connection**| Computers, terminals, and peripheral devices are typically physically connected with distinct cabling (Ethernet/fiber). | There may not be a direct hardware physical connection between the various endpoint computers. |
> | **Network Speed** | Much higher than WAN. The maximum speed of a typical LAN easily reaches 1000 Megabits per second (Gbps). | Slower than LAN due to lower bandwidth availability. Speeds often max out around 150 Mbps externally. |
> | **Transmission Errors**| Fewer data transmission errors due to controlled short-distance mediums. | Higher error rates due to complex routing over vast distances and varied providers. |

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Introduction_to_WANs.pdf, WAN_Technology_Overview.pdf

Feature	LAN	WAN
Coverage	Building/campus	State/country/global
Speed	1–100 Gbps	Typically ≤ 150 Mbps
Ownership	Organization-owned	Leased from ISPs/carriers
Cost	Low (internal cabling)	High (leased lines, satellites)
Error Rate	Very low	Higher due to distance

Q: Intranet vs Extranet

Asked: 2022 Q.1c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.1c Model Answer**
> **Definitions:**

- **Intranet:** An intranet is a private computer network that is contained purely within an enterprise. It may consist of many interlinked local area networks used securely by internal employees.
- **Extranet:** An extranet is a computer network that allows controlled access from the outside, for specific business or educational purposes. It is an extension of an organization's intranet extended to isolated users outside the organization.

Differences:

Feature	Intranet	Extranet
Primary Audience	Strictly internal employees and staff of the organization.	Trusted external parties (partners, vendors, suppliers, and customers).
Access Boundaries	Contained entirely within the enterprise firewalls.	Permits secure, isolated access penetrating the corporate perimeter from the public internet (often utilizing VPNs).
Purpose	Internal collaboration, HR portals, corporate data sharing.	Supply chain management, B2B procurement, joint ventures.
Security Mechanism	Relies on internal physical network security and login domains.	Heavily relies on access VPNs, tunneling (L2F, PPTP), and strict authentication to establish trust.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Enterprise_Networks_and_VPNs.pdf

- **Intranet:** Private network strictly for internal employees. Behind corporate firewalls. Uses standard web technologies (HTTP, email) but only accessible inside.
- **Extranet:** Extension of the intranet granting controlled access to external trusted partners/vendors/customers via VPN or secure login.

Q: Simplex, Half-Duplex, Full-Duplex Communication

Asked: 2022 Q.1b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.1b Model Answer**
> **Modes of Data Transmission:**
> Data transmission modes describe the direction of signal flow between two linked devices.

Differences among the Modes:

Feature	Simplex	Half-Duplex	Full-Duplex
Direction of Flow	Unidirectional (one-way only).	Bidirectional, but only **one direction at a time**.	Bidirectional **at the same time** simultaneously.
Channel Usage	Dedicated fully to the sender transmitting to the receiver.	The single communication channel is shared by both, taking turns to transmit.	The communication channel is used in both directions concurrently (often via two separate wires or frequency bands).
Performance	Lowest throughput.	Moderate throughput (wait times during turn-around).	Highest throughput (no waiting).
Examples	Radio broadcast, Television broadcasting, Keyboard to CPU.	Walkie-talkie communication.	Telephone line conversations, modern Ethernet switches (100BASE-T full duplex).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Data_Communication_Fundamentals.pdf

Mode	Direction	Example
Simplex	One-way only	TV broadcast, Keyboard→CPU
Half-Duplex	Two-way, one at a time	Walkie-talkie
Full-Duplex	Two-way, simultaneous	Telephone, modern Ethernet

Q: Centralized vs Distributed Networks

Asked: 2022 Q.2c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.2c Model Answer
Feature	Centralized Network	Distributed Network
Structure	A type of network where all users directly connect to a singular central server.	A computer network that is structurally spread over various different networks acting jointly or separately.
Role of Server	The central server acts as the absolute agent for *all* communications. It uniquely stores both communication records and all user account info.	Functionality and resources are physically decentralized. Multiple nodes/servers operate cooperatively across the network.
Processing	All heavy processing and computations are executed exclusively on the central machine.	Besides shared communication, a distributed network often actively distributes *processing power* across participating nodes.
Single Point of Failure	Highly vulnerable. If the central database/server falls, all users are affected and the network halts.	Highly robust. There is no single point of failure. If one machine fails, the network and cached data can reroute to others.
Scalability	Generally does not scale well gracefully beyond its hardware limits.	Scales easily by adding more nodes/servers dynamically to the distributed pool.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Centralized:** All users connect to one central server. Single point of failure. Simple but non-scalable.
- **Distributed:** Multiple cooperating servers/nodes. No single point of failure. Highly scalable and robust.

Q: Message Switching vs Packet Switching

Asked: 2022 Q.8a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.8a Model Answer
Message Switching vs. Packet Switching:

Feature	Message Switching	Packet Switching
Data Formatting	The entire message is transmitted from node to node intact as a single, large block of data.	The originating station breaks the long message structurally into smaller, fixed-size chunks called "packets".
Storage & Forwarding	Intermediate nodes must possess enough memory/disk space to temporarily store the entire massive message before forwarding it to the next hop.	Intermediate memory requirements are low (switching via fast system memory). Packets are independently buffered and swiftly forwarded one at a time to the network.
Transmission Delays	Suffers from very high latency and transmission delays since a node cannot begin sending the message to the next hop until the entire massive file is completely received.	Substantially reduces delay through pipelining. Node 2 can begin forwarding the first packet of a message while Node 1 is still transmitting the second packet.
Channel Occupancy	Dominates the shared link for an extended period, blocking other users.	Allows many pairs of nodes to interleave packets over the same shared channel simultaneously, drastically improving multiplexing efficiency.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Packet_Switching_in_WAN.pdf, Switched_Communication_Networks_Part_1.pdf

Feature	Message Switching	Packet Switching
Data Unit	Entire message as one block	Message broken into small packets
Storage	Intermediate nodes store full message (huge memory)	Nodes buffer small packets (fast memory)
Delay	Very high (must receive entire message before forwarding)	Low (pipelining — forward packets as they arrive)
Channel Usage	Blocks the link for the full message duration	Multiple users interleave packets on same link

---

Q: Frame Relay and ATM

Asked: 2022 Q.8b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.8b Model Answer
Disadvantages of ATM (Asynchronous Transfer Mode):

- The hardware cost of specialized ATM switching devices is incredibly high.
- Because it uses tiny fixed-length packets called "cells" (typically 53 bytes), the overhead generated by the 5-byte cell header is proportionally very high compared to the payload.
- The ATM Quality of Service (QoS) mechanism is technically quite complex to properly administer.

Disadvantages of Frame Relay:

- Provides strictly unreliable service.
- The exact order of arriving packets at the destination may not be maintained.
- If erroneous packets are detected, they are directly dropped without notification.

Is Frame Relay reliable?
No, Frame Relay is specifically built to be an **unreliable** network.
Reason: Frame relay does not offer any built-in windowing flow control to prevent congestion. Crucially, there is no provision for the acknowledgment of received packets at the data link layer, and zero mechanism for retransmission control of corrupted frames. It implicitly assumes the underlying physical fiber lines are extremely clean (error rate of 1 bit in 10^{14}) and pushes the entire burden of error correction end-to-end strictly to the upper transport layers (like TCP).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: atm datagram virtual ckt.pdf, X25_Virtual_Circuits_DTE_DCE_PSE.pdf

Frame Relay: Packet-switched WAN technology. **Unreliable by design** — no ACKs, no retransmission at Layer 2. Assumes clean fiber lines. Error recovery pushed to TCP. Simple and fast but drops erroneous frames silently.

ATM (Asynchronous Transfer Mode): Uses small fixed-size "cells" (53 bytes: 5-byte header + 48-byte payload). High overhead ratio. Excellent QoS but expensive hardware. Largely replaced by MPLS.

Q: EGP vs OSPF

Asked: 2022 Q.8c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.8c Model Answer
Differences between EGP and OSPF:

Feature	Exterior Gateway Protocol (EGP / BGP)	Open Shortest Path First (OSPF)
Routing Scope	Categorized as an Exterior Gateway Protocol.	Categorized as an Interior Gateway Protocol (IGP).
Operational Boundary	Specifically designed to route packets logically *between* entirely independent Autonomous Systems (AS's) on the global internet.	Specifically designed to route packets strictly *within* a single, localized Autonomous System using common internal metrics.
Underlying Algorithm	Usually based on generic Distance-Vector logic (Path Vector).	Uses complex Link-State routing algorithms and metrics based strictly on link costs.
Supernetting Support	Older EGP protocols do not support supernetting / CIDR aggregation directly, though its modern successor BGP does perfectly.	Fully supports CIDR and Supernetting out of the box, drastically simplifying massive internal routing tables.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Feature	EGP/BGP (Exterior)	OSPF (Interior)
Scope	Between Autonomous Systems	Within one AS
Algorithm	Path Vector (BGP)	Link State (Dijkstra)
Driven By	Policy, cost, agreements	Performance metrics (link cost)
CIDR Support	BGP fully supports	Fully supports

Q: Connection-Oriented vs Connectionless Service

Asked: 2023 Q.5a, 2022 Q.5a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2023 Q.5a Model Answer
Connection-oriented Service:

- **Definition:** Analogous to the telephone system. It requires communication entities to use a handshake process to establish a connection before sending data. TCP provides connection-oriented services.
- **Properties:** Transmits data as a continuous stream of bytes. Reliability is strictly achieved by having the recipient acknowledge each message. Packets are received in order due to sequencing and flow control. Uses virtual circuit switching. It is vulnerable to router failure along the dedicated path.
- **Application Examples:** File transfer (FTP), remote login (Telnet, SSH), and video on demand. These require guaranteed, ordered delivery.

Connection-less Service:

- **Definition:** Analogous to the postal system. Packets of data (datagrams) are transmitted from source to destination directly without establishing a connection. UDP and IP provide connectionless service.
- **Properties:** Handles packets individually. Each packet carries a full destination address. Packets do not follow a fixed path and can arrive out of order. It is robust to router failure since alternate paths can be taken dynamically.
- **Application Examples:** Credit card point-of-sale verification, electronic funds transfer, remote database access queries, DNS. These are inherently query/reply where speed and robustness are preferred over sustained stream overhead.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: Datagram_vs_Virtual_Circuit_Packet_Switching.pdf

Feature	Connection-Oriented (TCP)	Connectionless (UDP/IP)
Analogy	Telephone call	Postal system
Setup	3-way handshake required	No setup, send directly
Ordering	Guaranteed in-order delivery	May arrive out of order
Reliability	ACKs, retransmission	Best-effort, no guarantees
Robustness	Vulnerable to path failure	Routes around failures

Q: CRC with Generator $G(x) = x^5 + x + 1$

Asked: 2022 Q.2c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.2c Model Answer
Feature	Centralized Network	Distributed Network
Structure	A type of network where all users directly connect to a singular central server.	A computer network that is structurally spread over various different networks acting jointly or separately.
Role of Server	The central server acts as the absolute agent for *all* communications. It uniquely stores both communication records and all user account info.	Functionality and resources are physically decentralized. Multiple nodes/servers operate cooperatively across the network.
Processing	All heavy processing and computations are executed exclusively on the central machine.	Besides shared communication, a distributed network often actively distributes *processing power* across participating nodes.
Single Point of Failure	Highly vulnerable. If the central database/server falls, all users are affected and the network halts.	Highly robust. There is no single point of failure. If one machine fails, the network and cached data can reroute to others.
Scalability	Generally does not scale well gracefully beyond its hardware limits.	Scales easily by adding more nodes/servers dynamically to the distributed pool.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- Generator Binary: x^5+x+1 → `100011` (degree 5)
- Input frame: `101101110011`
- Append 5 zeros: `10110111001100000`
- Perform modulo-2 division by `100011`
- The 5-bit remainder is the CRC checksum
- Transmitted frame = original + remainder

Q: Noiseless Channel Bit Rate Calculation (Nyquist)

Asked: 2022 Q.2b

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2022 Q.2b Model Answer
Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Geographical Area	Restricted to a limited geographical coverage such as a house, single office building, or school campus.	Spans a large geographic area, such as a state, province, or country.
Data Transmission Cost	Much lower, since the data transmission medium is usually owned outright by the user organization.	Very high, because the transmission mediums used are leased lines or public carrier systems (telephone lines, microwaves, satellites).
Physical Connection	Computers, terminals, and peripheral devices are typically physically connected with distinct cabling (Ethernet/fiber).	There may not be a direct hardware physical connection between the various endpoint computers.
Network Speed	Much higher than WAN. The maximum speed of a typical LAN easily reaches 1000 Megabits per second (Gbps).	Slower than LAN due to lower bandwidth availability. Speeds often max out around 150 Mbps externally.
Transmission Errors	Fewer data transmission errors due to controlled short-distance mediums.	Higher error rates due to complex routing over vast distances and varied providers.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Problem: Signal through noiseless channel, capacity 128 Kbps max.
Using Nyquist: $C = 2 \times B \times \log_2(L)$.
If binary (L=2): $128000 = 2 \times B \times 1$ → **B = 64,000 Hz = 64 KHz**

Q: HTTP vs HTTPS, Telnet vs SSL, FTP vs HTTP, Port Numbers

Asked: 2019 Q.3a–d

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

Source: 2019 Q.3a Model Answer

- Data: `10011011100`
- Generator: x^4+x^2+1 → Binary: `10101` (degree 4)
- Augment data with 4 zeros: `100110111000000`

Modulo-2 Division:

```
             10000111001
          ________________
   10101 | 100110111000000
           10101
           -----
             01101
             00000
             -----
              11011
              10101
              -----
               11101
               10101
               -----
                10000
                10101
                -----
                 01010
                 00000
                 -----
                  10100
                  10101
                  -----
                   00010
                   00000
                   -----
                    00100
                    00000
                    -----
                     0100 (Remainder)
```

- **CRC Remainder:** `0100`
- **Transmitted Codeword:** `100110111000100`

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: HTTP_Fundamentals.pdf, Hyper_Text_Transfer_Protocol_HTTP.pdf

Protocol	Port	Description
HTTP	80	Unencrypted web page transfer
HTTPS	443	HTTP + TLS/SSL encryption
FTP	20/21	File transfer (data/control channels)
Telnet	23	Unencrypted remote terminal (insecure)
SSH	22	Encrypted remote terminal (replaces Telnet)
SMTP	25	Sending email
DNS	53	Domain name resolution
POP3	110	Downloading email
IMAP	143	Syncing email

- **HTTP vs HTTPS:** HTTPS adds TLS/SSL encryption layer. Validates server identity via certificates.
- **Telnet vs SSH:** Both provide remote terminal. Telnet sends plaintext passwords; SSH encrypts everything.
- **FTP vs HTTP:** FTP optimized for bulk file transfer (resume, binary mode). HTTP optimized for stateless web page requests.

Q: IP Datagram Format (IPv4 Header)

Asked: 2021 Q.6d

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2021 Q.6d Model Answer**
> **ACL (Access Control List):**
> *(Detailed extensively in 2022 Q.8a)* A strict firewalling sequence embedded physically on router ports filtering IP/Ports deterministically.

IP Datagram Format Structurally:
Depending on the version, the structure changes:

- **IPv4 Datagram:** Optimized 20-byte core header containing: Version (4 bits), IHL (4 bits), Type of Service (8 bits for QoS), Total Length (16 bits), properties for fragmentation (Identification, Flags, Fragment Offset), TTL, Protocol (8 bits), Header Checksum (16 bits), and Source/Destination IPs (32 bits each).
- **IPv6 Datagram:** Fixed-length 40-byte base header. The fixed format helps speed processing. Includes: Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Source/Destination IPs (128 bits each). Unlike IPv4, no fragmentation is allowed by routers (handled exclusively by endpoints).

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 99 | Slides: Internet_Protocol_Router_Forwarding.pdf

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|Version|  IHL  |Type of Service|          Total Length         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|         Identification        |Flags|     Fragment Offset     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Time to Live |    Protocol   |       Header Checksum         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       Source IP Address                       |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Destination IP Address                     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Minimum header = 20 bytes. **TTL** = max hops before discard. **Protocol** = upper layer (TCP=6, UDP=17).

Q: ACL vs Firewall

Asked: 2020 Q.5c/d

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2020 Q.5c Model Answer**
> *(Route poisoning & Split horizon meticulously detailed in 2021 Q.6b).*

ACL vs. Firewall:

- **ACL (Access Control List):** A mathematically ordered sequence of deterministic packet-filtering rules statically configured dynamically on router interfaces. It filters specifically based on Layer 3 (IPs) and Layer 4 (Ports). It structurally is stateless, blindly examining each packet in utter isolation purely against the rule list.
- **Firewall:** A distinct, vastly more powerful massive hardware/software security appliance (or server cluster). Firewalls natively implement ACLs, but vastly extend them securely to Layer 7 (Deep Packet Inspection). They are crucially **stateful** (dynamically tracking the state of massive, multi-step TCP application connections). They structurally protect two networks by definitively sitting explicitly between the Corporate Network Premises and the external Internet.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **ACL:** Simple ordered rule list on router interfaces. Stateless — examines each packet in isolation. Filters Layer 3-4 (IP/Port).
- **Firewall:** Full security appliance. **Stateful** — tracks TCP connection state. Deep Packet Inspection (Layer 7). Sits between internal network and internet. Implements ACLs as one of many features.

Q: Flow Control and Error Control Methods

Asked: 2019 Q.4d
Rifat PDF: Pages 54–55 | Slides: Data_Link_Flow_and_Error_Control.pdf

Flow Control: Regulates sender speed to prevent overwhelming the receiver.

- **Stop-and-Wait:** Send 1 frame, wait for ACK. Simple but wasteful.
- **Sliding Window:** Send W frames before waiting. Efficient.

Error Control: Detects and corrects errors.

- **Parity Check:** Single bit added. Detects odd number of errors.
- **CRC:** Polynomial division. Detects burst errors.
- **Hamming Code:** Locates and corrects single-bit errors.

Q: 127.0.0.0 (Loopback Address)

Asked: 2019 Q.4a

127.0.0.0/8 is the loopback address block. 127.0.0.1 is used to test the local TCP/IP stack without sending packets onto the actual network. Any packet sent to this address is looped back internally by the OS.

PART I: FINAL AUDIT — REMAINING GAPS

Q: TCP Connection Termination (4-Way Handshake)

Asked: 2022 Q.3a (terminate)

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2022 Q.3a Model Answer**
> **TCP (Transmission Control Protocol):**
> TCP is the primary transport level protocol of the Internet Protocol Suite that provides reliable, byte-oriented stream management. It offers a connection-oriented service where applications interact with TCP to send and receive data reliably across an IP network with sequencing and flow control.

TCP 3-Way Handshake Process (Connection Establishment):
The reliable handshake ensures both communication entities establish parameters before sending data.

1. **SYN (Step 1):** The Active Participant (Client) sends a segment with the SYN (Synchronize) flag set to request a connection to the Passive Participant (Server). It includes the client's initial Sequence number `X`.
2. **SYN-ACK (Step 2):** The Server receives the SYN, allocates data structures, and replies with a segment that has *both* the SYN flag and ACK flag set. The Server provides its own initial Sequence number `Y`, and acknowledges the client's sequence by setting the Acknowledgment number to `X + 1`.
3. **ACK (Step 3):** The Client receives the SYN-ACK, and sends a final segment with just the ACK flag set, acknowledging the server's sequence by setting the Acknowledgment number to `Y + 1`.

(See diagram in the final document: TCP 3-Way Handshake)

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 119–120 | Slides: Transmission_Control_Protocol_TCP.pdf

TCP connection is **terminated** using a 4-way handshake (FIN exchange):

1. **FIN (Client→Server):** Client sends FIN segment, enters FIN-WAIT-1.
2. **ACK (Server→Client):** Server acknowledges FIN, enters CLOSE-WAIT. Client enters FIN-WAIT-2.
3. **FIN (Server→Client):** Server finishes sending data, sends its own FIN.
4. **ACK (Client→Server):** Client acknowledges Server's FIN, enters TIME-WAIT (waits 2×MSL), then CLOSED.

Q: Routing vs Forwarding

Asked: 2021 Q.5a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2021 Q.5a Model Answer**
> > **Two Major Classes of Routing Algorithms:**
> > The two foundational classes are **Distance Vector (DV)** routing (which utilizes fundamentally distributed Bellman-Ford calculations based exclusively on neighbor rumors, like RIP) and **Link State (LS)** routing (where every node independently maps the entire global topology mathematically using Dijkstra's algorithm, like OSPF).
>
> **Routing vs. Forwarding:**
>
> - **Routing:** The immensely complex "control plane" process. It spans the entire globally distributed network to mathematically calculate and determine strictly the optimal end-to-end paths from a source to a specific destination. It builds the routing tables.
> - **Forwarding:** The exceptionally fast, localized "data plane" action. When a singular packet biologically arrives at a router's input port, the router does a rapid hardware lookup in its forwarding table and dynamically switches the packet instantly to the appropriate output physical port.
>
> **Why we need Distance Vector and Link State:**
> We need distinct protocols because networks scale vastly differently. **Distance Vector** is fiercely needed for exceedingly simple, small-to-medium networks because it requires minuscule memory overhead and almost zero CPU processing power. **Link State** is absolutely critical for massive enterprise/ISP networks because, despite needing high CPU/RAM to hold the topological map, it completely avoids deadly routing loops, scales infinitely better, and converges mathematically instantly upon link failures.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 78, 81

- **Routing (Control Plane):** The global, distributed process of calculating optimal end-to-end paths across the entire network. Builds routing tables. Runs algorithms like Dijkstra or Bellman-Ford.
- **Forwarding (Data Plane):** The fast, local action at a single router. When a packet arrives, the router looks up the destination IP in its forwarding table and switches the packet to the correct output port. Happens in hardware at wire speed.

Q: Link State Routing Algorithm — Steps

Asked: 2022 Q.7a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2022 Q.7a Model Answer**
>
> - **Flow Control:** This is the detailed mechanism of adjusting the specific sending rate according strictly to the resources (e.g., buffer size) currently available at the final *destination* receiver.
> - **Congestion Control:** This defines algorithms and protocols that attempt to prevent or resolve the over-subscription of *network* resources (specifically links and routers) by dialing back the overall volume of packets injected into the network.
>
> **Why Congestion Control Differs from Flow Control:**
> They are frequently mixed up, but their scope of regulation is drastically different.
>
> - Flow control is strictly "point-to-point." It is limited in a way that it only considers what is going on exclusively between the two end devices (Sender A and Receiver B). Its sole goal is ensuring the sender does not overwhelm the receiver's memory buffers.
> - Congestion control is a "global" network issue. It regulates traffic to ensure the sender does not overwhelm the *intermediate routers and switches* forming the public network fabric.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 84–86 | Slides: Routing_Implementation_Part_2.pdf

Steps of Link State Routing:

1. **Discover Neighbors:** Each router discovers its directly connected neighbors and their link costs.
2. **Build LSP (Link State Packet):** Each router constructs a packet containing its identity, sequence number, age, and list of neighbors with costs.
3. **Flood LSP:** Each router distributes its LSP to ALL other routers in the network (not just neighbors).
4. **Build Topological Map:** Each router independently assembles a complete graph of the entire network from all received LSPs.
5. **Run Dijkstra:** Each router independently computes the shortest path tree from itself to every other router using Dijkstra's algorithm.

Q: Prominent Factors/Causes of Congestion

Asked: 2023 Q.6c, 2020 Q.6a, 2019 Q.6c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.6c Model Answer**
> **Superiority of Token Bucket:**
> The **Token Bucket algorithm** provides better overall network performance and flexibility compared to the Leaky Bucket algorithm.

Why (Based on Implementation Differences):

- **Handling Burstiness:** An important difference is that Leaky Bucket enforces a very rigid pattern on the output stream, discarding packets immediately when the bucket is full. It forces bursty traffic to smooth out entirely into a fixed, constant rate.
- **Permitting Capacity:** Token bucket is less restrictive. It permits burstiness but bounds it. It collects tokens generated at a regular rate. Packets waiting in a queue must grab a token before being transmitted. If a sudden burst of data arrives and the bucket has saved up tokens, those bursts of packets can be sent all at once.
- **Discard Policy:** Token bucket throws away excess *tokens* when the bucket is full, but never drops the actual user *packets* (they simply wait in queue until new tokens generate). Leaky bucket drops *packets* when full. Thus, token bucket adapts far better to the bursty nature of modern web traffic without causing data loss.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 88–89 | Slides: Congestion_Control_in_Routing.pdf

Congestion occurs when total packets injected exceed network carrying capacity:

1. **Insufficient bandwidth:** Too many hosts pumping data beyond link capacity.
2. **Slow router processors:** Slow switching fabrics bottleneck queued packets.
3. **Inadequate buffer memory:** Router queues overflow, causing packet drops.
4. **Mismatched line speeds:** A fast LAN dumping into a slow WAN.
5. **Bursty traffic patterns:** Multiple hosts burst simultaneously.

Q: UTP vs STP (Twisted Pair)

Asked: 2023 Q.1b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.1b Model Answer**
> **Advantages of UTP over STP:**

- **Cost-Effectiveness:** UTP is significantly less expensive to manufacture and purchase than STP because it lacks the internal foil or braided shielding.
- **Installation & Flexibility:** UTP has a much smaller outer diameter and is more flexible, making it easier to install, route through tight conduits, and terminate than the bulkier STP cable.
- **Maintenance:** UTP requires less maintenance. STP shielding must be properly grounded at both ends; if grounding is imperfect, the shield acts as an antenna, drawing in more noise than it blocks. UTP eliminates this complex grounding requirement.
- **Adequate for Most Usage:** For most indoor, standard LAN topologies (Voice grade Cat 3, Data grade Cat 5 up to 100 Mbps), UTP provides perfectly adequate transmission without the added bulk and cost of STP, especially in Star topologies.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 58–59 | Slides: Physical_Layer_Transmission_Media.pdf

- **UTP (Unshielded Twisted Pair):** Cheaper, thinner, easier to install. Adequate for most indoor LANs (Cat5e/Cat6). No grounding required.
- **STP (Shielded Twisted Pair):** Has foil/braided shielding. Better EMI protection for industrial environments. More expensive, bulkier, must be properly grounded (improper grounding makes shield act as antenna).

Q: Star-Ring Hybrid Topology

Asked: 2023 Q.1c

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.1c Model Answer**
> **Hybridizing Star and Ring Topologies:**
> When these topologies are hybridized, the physical network is wired like a **Star** (with all nodes connecting to a central hub/coupler), but logically the data travels in a **Ring** path. The central device acts as a Multistation Access Unit (MAU) that internally bridges the transmit pair of one node to the receive pair of the next node.

Circumstances and Challenges in Setup:

- **Cabling Medium:** Fiber optics or twisted pair are suitable for use in ring and star topologies. Setting this up requires running point-to-point links (one for transmit, one for receive) from every node back to the central hub.
- **Central Point of Failure:** While the logical ring prevents packet collisions, the physical setup creates a single point of failure at the central hub. If the internal ring wiring in the MAU fails, the entire network drops.
- **Fault Detection:** Setting up the network requires hardware that can automatically bypass a faulty node or broken cable run, allowing the ring to instantly close loop and preserve the remaining connectivity.

(See diagram in the final document: Network Topologies)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Slides: LAN_Technology_and_Network_Topologies.pdf

Hybridizing Star + Ring: Physically wired as a Star (nodes connect to central MAU — Multistation Access Unit), but logically data flows in a Ring. The MAU internally connects each node's transmit to the next node's receive.

Setup Challenges: (1) Central MAU = single point of failure. (2) Need hardware to auto-bypass faulty nodes. (3) Requires separate TX/RX pairs from each node to hub.

Q: Extended Star Topology vs Standard Star

Asked: 2023 Q.2a

[!NOTE] 🕮 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.2a Model Answer**
> **Extended Star Topology vs. Standard Star Topology:**

- **Network Expansion:** In a standard star topology, each station directly connects to a single common central node (star coupler). An extended star topology connects individual star central nodes to a higher-level central node, building a hierarchical structure.
- **Distance/Reach:** An extended star allows the network to cover a much larger geographical area (like spanning multiple floors or buildings) by distributing hubs, whereas a standard star is limited by the maximum cable length from the single central hub.
- **Compartmentalization:** It groups nodes logically, helping contain local traffic within local hubs without flooding the core central hub.

Disadvantages of Star Topology:

- **Single Point of Failure:** The entire network depends entirely on the central node/coupler. If the central hub fails, all connected stations lose connectivity.
- **Cable Cost (Proportional to 'n'):** Every single station must have its own dedicated physical cable span running all the way back to the central node. As the number of links (n) grows, wiring complexity and cabling cost increase rapidly compared to a Bus topology.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

- **Standard Star:** All nodes connect to ONE central hub/switch.
- **Extended Star:** Multiple star sub-networks connected hierarchically via a backbone. Increases coverage area and compartmentalizes traffic.
- **Disadvantages of Star:** (1) Central point of failure. (2) Cable cost proportional to number of nodes.

Q: Client-Server vs Peer-to-Peer Model; Computer vs Autonomous System

Asked: 2023 Q.4a, 2022 Q.1a, 2021 Q.1a

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.4a Model Answer**
> **IP Address Classes and Ranges:**
> Traditional IPv4 addresses are divided into classes to accommodate networks of varying sizes.

- **Class A:** `0.0.0.0` to `127.255.255.255`. Designed for extremely large networks.
- **Class B:** `128.0.0.0` to `191.255.255.255`. Designed for medium-sized networks.
- **Class C:** `192.0.0.0` to `223.255.255.255`. Designed for smaller local area networks.
- **Class D:** `224.0.0.0` to `239.255.255.255`. Reserved strictly for Multicast addressing.
- **Class E:** `240.0.0.0` to `255.255.255.255`. Reserved for experimental/future use.

Components to Determine Identifiers:
An IP address is logically broken into two primary components, separated using a subnet mask or CIDR notation:

1. **Network Prefix (Network ID):** A common group of most-significant bits used for routing datagrams to the correct specific network.
2. **Rest Field (Host ID):** The remaining bits used to uniquely identify a specific host/node within that localized network.

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 1–3

Client-Server: Centralized. Thin clients request from powerful servers. Easy to manage. Examples: Web browsing, Email, Cloud apps.
Peer-to-Peer (P2P): Decentralized. Every node acts as both client and server. No central dependency. Examples: BitTorrent, Bitcoin.

Computer vs Autonomous System (AS):

- **Computer:** A single end-device (host) on a network.
- **Autonomous System (AS):** A set of routers under single administrative control, using one IGP internally, presenting a unified routing policy externally via BGP.

Q: IPv4 Packet Fragmentation Calculation (MTU = 500 bytes)

Asked: 2023 Q.8c

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.8c Model Answer**
> **Functional Differences:**

- **Standard ACL:** Can act only based on the **Source IP** address. It either blocks or permits all traffic from that source entirely. Because it does not check destinations, it should generally be placed as close to the destination router as possible to avoid dropping traffic intended for other locations unnecessarily.
- **Extended ACL:** highly granular. It can filter traffic based on **Source IP, Destination IP, Protocol (TCP, UDP, ICMP), and Port Numbers**. Because it is highly specific, these should be placed as close to the source router as possible to block unwanted traffic before it consumes bandwidth traversing the network.

Basic Configuration Commands (Cisco syntax):

- **Standard ACL (Number range 1-99):**
  `access-list 10 deny host 192.168.1.5`
  `access-list 10 permit any`
  (To apply): `ip access-group 10 in`
- **Extended ACL (Number range 100-199):**
  `access-list 110 permit tcp 192.168.0.0 0.0.0.255 host 10.0.0.5 eq 80`

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Page 99 | Slides: Network_Season_10_Fragmentation.pdf

Method for fragmenting an IPv4 packet when outgoing MTU = 500 bytes:

1. **Max data per fragment** = MTU - IP Header = 500 - 20 = 480 bytes.
2. **Fragment offset unit** = 8 bytes. So max data must be a multiple of 8: 480 ÷ 8 = 60 (exact, good).
3. **Number of fragments** = ⌈Total Data / 480⌉
4. Each fragment gets: same Identification field, MF (More Fragments) flag set on all but the last, Fragment Offset = (previous fragments' total data) / 8.

Example: Original packet = 4000 bytes total (3980 data + 20 header).

- Fragment 1: Offset=0, Data=480 bytes, MF=1
- Fragment 2: Offset=60, Data=480 bytes, MF=1
- ... continue until all data is covered
- Last Fragment: MF=0

Q: Why Do We Need Cryptography?

Asked: 2023 Q.7b

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.7b Model Answer**
> > **Virtual Circuit Packet Switching:**
>
> - Operates like circuit switching dynamically. Before data flows, a virtual path is established edge-to-edge.
> - Each packet contains a Virtual Circuit Identifier (VCI) in its header.
> - **Routing Decision:** No complex routing decisions are required at each node; routers simply look up the VCI and forward along the pre-established path.
> - Provides Quality of Service (QoS), guaranteed sequential delivery, and minimal per-packet processing delay, but is highly vulnerable to link failure (the whole circuit drops).
>
> **Datagram Packet Switching:**
>
> - Operates purely connectionless. Every packet carries full source and destination IP addresses.
> - **Routing Decision:** Routing decisions are deeply made dynamically for each individual datagram at every single router hop.
> - Datagrams can follow entirely different routes to the destination and arrive out of order.
> - Highly robust. If a router malfunctions, datagrams simply route around the failure dynamically.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 129, 134

Reasons:

1. **Confidentiality:** Prevent unauthorized reading of sensitive data.
2. **Integrity:** Detect if data has been tampered with during transmission.
3. **Authentication:** Verify the identity of the sender.
4. **Non-repudiation:** Sender cannot deny having sent a message (via digital signatures).

Q: "Explicitly Deny" in ACL

Asked: 2023 Q.7a

> [!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**
>
> > **Source: 2023 Q.7a Model Answer**
>
> - **Flow Control:** This is a vital mechanism regulating the transmission of data strictly between two end devices (Sender and Receiver). It considers only what is going on within that specific session. Its primary goal is to ensure a fast sender does not overwhelm a slow receiver. It is managed by advertising a receive window size.
> - **Congestion Control:** This operates globally. It regulates the flow of packets coming into the network as a whole, preventing the entire network fabric (routers and links) from becoming overloaded. Its goal is to prevent network saturation, dropped packets, and transmission delays across all users.
> - *Interaction:* In TCP, the Effective Window Size (the actual amount of data the source can send) is dictated by taking the mathematically stricter of the two: `MIN(CongestionWindow, AdvertisedWindow)`.

> [!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Every ACL has an **implicit deny all** at the end — any packet that doesn't match any rule is silently dropped. An **explicit deny** is when the administrator deliberately writes a deny rule (e.g., `access-list 10 deny host 192.168.1.100`) to specifically block certain traffic and optionally log it, making the denial visible and auditable rather than silent.

Q: DHCP Leasing and Renewal

Asked: 2023 Q.8b

[!NOTE] 📖 **UNIVERSITY EXAM (DETAILED PROFESSOR-LEVEL ANSWER):**

> **Source: 2023 Q.8b Model Answer**
> **Why DHCP Process is Needed:**
> Dynamic Host Configuration Protocol (DHCP) automatically provides vital network configuration information (IP address, subnet mask, gateway address) to hosts on lease. Connecting a host manually is highly prone to mistakes (e.g., administrator assigns an already used IP address causing IP conflicts). DHCP dynamically distributes these parameters perfectly, allowing devices to freely log on, move between networks, and request temporary configuration without any manual human intervention.

DHCP DORA Process:

- **D - DHCPDISCOVER:** A new node connects and broadcasts (to `255.255.255.255`) requesting an IP. Source IP is `0.0.0.0`.
- **O - DHCPOFFER:** The DHCP Server picks an available IP and unicassts/broadcasts a proposed offer (IP, lease time) back to the client.
- **R - DHCPREQUEST:** The host selects the offer and officially broadcasts an acceptance to the network to claim it.
- **A - DHCPACK:** The Server officially acknowledges the claim and marks the IP as unavailable in its storage.

(See diagram in the final document: DHCP DORA Process)

[!TIP] 🚀 **QUICK REVISION / SHORT NOTES:**

Rifat PDF: Pages 126–127 | Slides: DHCP_Protocol_Breakdown.pdf

- **DHCP Lease:** An IP address is assigned for a limited time period (lease time), not permanently. This ensures addresses return to the pool when devices disconnect.
- **DHCP Renewal:** At 50% of lease time (T1), client sends DHCPREQUEST to renew. If no response, at 87.5% (T2) it broadcasts to any DHCP server. If lease expires without renewal, client must restart the DORA process.

END OF WRITTEN ANSWERS — Diagram Appendix follows below.

📐 APPENDIX: MUST-KNOW DIAGRAMS (Hand-Draw These in Exam)

Every diagram below has been asked in exams. Practice sketching them.
Page numbers refer to the Rifat PDF. Slide files are in `materials/slides/`.

Diagram 1: Network Topologies

When asked: 2023 Q.1c, 2022 Q.2a, 2021 Q.1b, 2020 Q.1b | **Rifat PDF: Pages 11, 14–15**
Slide: `Network_Topology_Overview.pdf`, `LAN_Technology_and_Network_Topologies.pdf`

```
BUS:                  STAR:                   RING:
 ═══┬═══┬═══┬═══      [PC]  [PC]             [PC]──[PC]
    │   │   │           \   |   /                |       |
  [PC] [PC] [PC]        [HUB]                 [PC]──[PC]
                         /   |   \
                      [PC]  [PC]  [PC]

MESH (Full):           TREE/EXTENDED STAR:
[PC]──[PC]                 [Core Switch]
 |  ╲╱  |                 /     |      \
[PC]──[PC]            [Switch] [Switch] [Switch]
                      / | \    / | \     / | \
                    PCs PCs  PCs PCs   PCs PCs
```

Diagram 2: OSI vs TCP/IP Layers

When asked: 2023 Q.1a, 2020 Q.2a, 2019 Q.1a | **Rifat PDF: Pages 1, 4**
Slide: `OSI_and_TCP_IP_Layering_Models.pdf`

```
OSI Model (7)          TCP/IP Model (4)
┌──────────────┐       ┌──────────────┐
│ Application  │       │              │
├──────────────┤       │ Application  │
│ Presentation │       │              │
├──────────────┤       │              │
│   Session    │       │              │
├──────────────┤       ├──────────────┤
│  Transport   │       │  Transport   │
├──────────────┤       ├──────────────┤
│   Network    │       │   Internet   │
├──────────────┤       ├──────────────┤
│  Data Link   │       │              │
├──────────────┤       │Network Access│
│   Physical   │       │              │
└──────────────┘       └──────────────┘
```

Diagram 3: Encapsulation Process

When asked: 2023 Q.1a, 2022 Q.2a | **Rifat PDF: Pages 2–4**
Slide: `Introduction_to_Layering.pdf`

```
Application:  [      DATA      ]
Transport:    [TCP Hdr][  DATA  ]            ← Segment
Network:      [IP Hdr][TCP Hdr][DATA]        ← Packet
Data Link:    [MAC][IP Hdr][TCP][DATA][CRC]   ← Frame
Physical:     101010110011100101...            ← Bits
```

Diagram 4: TCP 3-Way Handshake

When asked: 2022 Q.3a, 2021 Q.3a, 2020 Q.2c | **Rifat PDF: Pages 119–120**
Slide: `Transmission_Control_Protocol_TCP.pdf`

```
Client                          Server
  |                                |
  |──── SYN (seq=X) ──────────────>|
  |                                |
  |<──── SYN-ACK (seq=Y, ack=X+1)─|
  |                                |
  |──── ACK (ack=Y+1) ────────────>|
  |                                |
  |====== CONNECTION ESTABLISHED ==|
```

Diagram 5: DHCP DORA Process

When asked: 2023 Q.8b | **Rifat PDF: Pages 126–127**
Slide: `DHCP_Protocol_Breakdown.pdf`, `DHCP_and_DNS_Lecture_52.pdf`

```
Client                            DHCP Server
  |                                    |
  |── (1) DHCP DISCOVER (broadcast) ->|
  |                                    |
  |<── (2) DHCP OFFER (IP proposed) ──|
  |                                    |
  |── (3) DHCP REQUEST (accept) ─────>|
  |                                    |
  |<── (4) DHCP ACK (confirmed) ──────|
  |                                    |
  |   [Client now uses assigned IP]    |
```

Diagram 6: ARP Process

When asked: 2022 Q.6a | **Rifat PDF: Page 3**
Slide: `Address_Resolution_Protocol_ARP.pdf`, `arp lec51.pdf`

```
Host A (knows IP, needs MAC)            Host B (target)
   |                                        |
   |── ARP Request (broadcast) ──────────> | (all hosts)
   |   "Who has 192.168.1.5?"              |
   |                                        |
   |<── ARP Reply (unicast) ───────────────|
   |   "I am 192.168.1.5, MAC=AA:BB:..."   |
   |                                        |
   [A caches B's MAC in ARP table]
```

Diagram 7: DNS Resolver Lookup

When asked: 2021 Q.7a, 2020 Q.7a | **Rifat PDF: Pages 123, 126**
Slide: `DNS_Hierarchical_Architecture.pdf`, `DNS_Lookup_Caching_DHCP_ICMP.pdf`

```
[Client PC] ──> [Local DNS Server]
                     |
              (1)    |──> [Root DNS Server (.)]
              (2)    |<── "Go ask .bd TLD server"
              (3)    |──> [.bd TLD Server]
              (4)    |<── "Go ask .ac.bd server"
              (5)    |──> [.ac.bd Authoritative]
              (6)    |<── "cuet.ac.bd = 203.0.113.5"
                     |
[Client PC] <── [Returns IP + Caches it]
```

Diagram 8: NAT Translation

When asked: 2023 Q.5b, 2022 Q.6b | **Rifat PDF: Pages 89, 94**
Slide: `Network_Address_Translation_NAT.pdf`

```
Private Network              NAT Router               Internet
┌──────────────┐         ┌──────────────────┐      ┌──────────┐
│192.168.1.50  │──────>  │ Replaces src IP  │──────>│          │
│   :3211      │         │ 192.168.1.50:3211│      │ Web      │
└──────────────┘         │ with             │      │ Server   │
                         │ 203.0.113.5:60001│      │104.21.3.4│
┌──────────────┐         │                  │      │          │
│192.168.1.65  │──────>  │ Translation Table│      └──────────┘
│   :5432      │         │ maps both ways   │
└──────────────┘         └──────────────────┘
```

Diagram 9: IPv6 Tunneling Through IPv4

When asked: 2022 Q.5b | **Rifat PDF: Page 97**
Slide: `IPv6_Extension_Headers_and_Tunneling.pdf`

```
[IPv6 Host] ──> [Dual-Stack Router A] ═══════════> [Dual-Stack Router B] ──> [IPv6 Host]
                       |        IPv4 Network              |
                       |                                  |
                  Encapsulate:                       Decapsulate:
                  [IPv4 Hdr][IPv6 Pkt]          Remove IPv4 Hdr
                  Protocol Field = 41           Deliver IPv6 Pkt
```

Diagram 10: Leaky Bucket vs Token Bucket

When asked: 2023 Q.6c, 2021 Q.5b | **Rifat PDF: Pages 88, 92, 142**
Slide: `Token_Bucket_vs_Leaky_Bucket.pdf`, `QoS_Leaky_Token_Bucket_Loadshedding.pdf`

```
LEAKY BUCKET:                    TOKEN BUCKET:
 Bursty Input                     Bursty Input
     ↓                                ↓
 ┌────────┐                       ┌───────┐
 │ Bucket │ ← overflow=DROP       │ Queue │ ← packets wait
 │        │                       └───┬───┘
 └───┬────┘                           │ need token to send
     │ fixed rate output              ┌───┴───┐
     ↓                              │ Token │ ← tokens added
 ═══════                            │ Bucket│   at fixed rate
 Constant                           └───┬───┘
 Rate                                   │
                                        ↓
                                  Bursty OK (up to bucket size)
```

Diagram 11: ACL Flowchart

When asked: 2022 Q.8a | **Rifat PDF: Pages 134–137**
Slide: `Firewalls_and_Virtual_Private_Networks.pdf`

```
Incoming Packet
     │
     ▼
┌──────────────┐      YES      ┌──────────┐
│ Match Rule 1?│──────────────> │ Permit/  │
└──────────────┘               │ Deny     │
       │                       └──────────┘
       │ NO
       ▼
┌──────────────┐      YES      ┌──────────┐
│ Match Rule 2?│──────────────> │ Permit/  │
└──────────────┘               │ Deny     │
       │                       └──────────┘
       │ NO
       ▼
┌──────────────┐
│ Implicit Deny All│  ← DEFAULT: drop packet
└──────────────┘

Standard ACL: Place near DESTINATION
Extended ACL: Place near SOURCE
```

Diagram 12: Transparent vs Non-Transparent Fragmentation

When asked: 2021 Q.4c, 2019 Q.8a | **Rifat PDF: Page 99**
Slide: `Transparent_vs_NonTransparent_Fragmentation.pdf`, `Network_Season_10_Fragmentation.pdf`

```
TRANSPARENT:
[Src]──>[G1: Fragment]──>[Small MTU Net]──>[G2: Reassemble]──>[Dest]
                                          ^ reassembled HERE

NON-TRANSPARENT (IPv4):
[Src]──>[Router: Fragment]──>[Net A]──>[Net B]──>[Dest: Reassemble]
                                                ^ reassembled HERE only
```

Diagram 13: Transposition Cipher Matrix (Column Method)

When asked: 2023, 2021, 2020, 2019 | **Rifat PDF: Page 138**

```
Key:  D  I  C  T  I  O  N  A  R  Y
      1  2  3  4  5  6  7  8  9  10
Row1: h  a  p  p  i  n  e  s  s  i
Row2: s  t  e  m  p  o  r  a  r  y

Step: Sort key alphabetically → read columns in that order.
A=col8, C=col3, D=col1, I=col2, I=col5, N=col7, O=col6, R=col9, T=col4, Y=col10
```

Diagram 14: FDDI Dual Ring

When asked: 2021 Q.1d, 2019 Q.8b | **Rifat PDF: Page 73**
Slide: `LAN_Technologies_Must_Read.pdf`

```
        ┌──[Station A]──┐
        │   ↓ Primary   │
   [Station D]      [Station B]
        │   ↑ Secondary │
        └──[Station C]──┘

If cable breaks between A & B:
Rings "wrap" → Primary + Secondary form one big loop
```

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